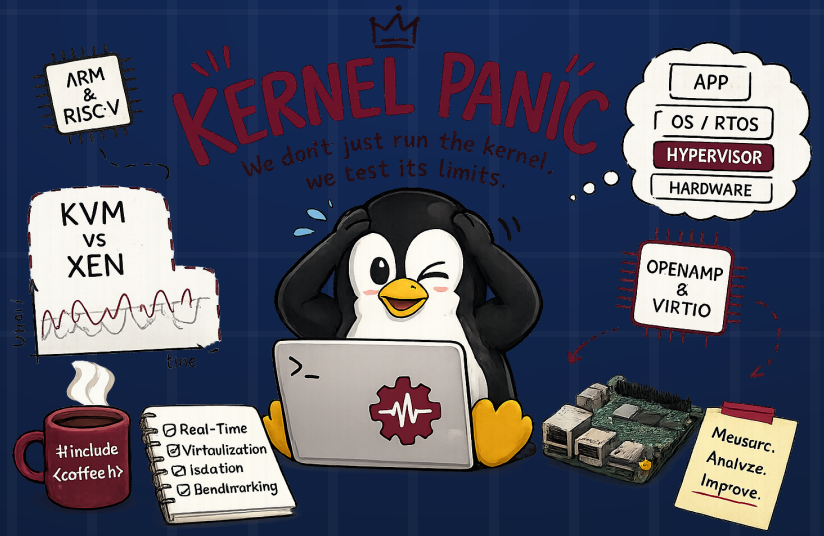
A6 PROJECT · AY 2025–26

Xen vs KVM under Mixed-Criticality Workloads

An Experimental Characterization



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COURSE
Real-Time Systems and Industrial Applications

OUTLINE

Agenda

① Motivation & Background

- Safety-critical systems & hypervisors
- Related work (Abeni & Faggioli 2019/2020)
- Our contribution

② Experimental Setup

- Testbed hardware & kernel configurations
- OS-level isolation techniques
- Measurement tools: cyclictest, TACLe, stress-ng

③ KVM Analysis

- Baseline, stress, isolation
- TACLe benchmarks (per-benchmark + summary)

④ Xen Analysis

- Credit2 vs vCPU Pinning
- Device Model Priority Inversion
- 32-scenario stressor study
- Noisy DomU study
- TACLe benchmarks & Dom0 stress
- PV vs PVH vs HVM

⑤ Conclusions

- KVM vs Xen head-to-head
- Key takeaways
- Open questions & future work
- References

01

Motivation & Background

Why do we need deterministic hypervisors — and what does the prior art say?

MOTIVATION

The Problem: Hard Real-Time in Virtual Machines

Safety-Critical Systems Need Bounded Latency

- Aerospace, automotive, industrial control — require strict WCET guarantees
- Consolidating workloads on shared hypervisors reduces cost & complexity
- **Problem:** co-located guests can temporally interfere with the real-time guest
- A single missed deadline can be catastrophic in safety-critical contexts

Two Hypervisor Architectures — Different Trade-offs

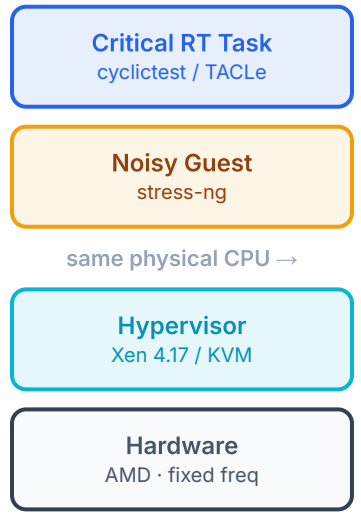
Xen (Type-1)

Bare-metal hypervisor. Dom0 control domain. HVM / PV / PVH guests. Direct hardware access.

KVM (Type-2)

Hosted hypervisor in Linux. Kernel module. Full hardware emulation via QEMU. PREEMPT_RT-compatible host.

🎯 **Key question:** Can commodity hypervisors guarantee hard real-time bounds for a mixed-criticality guest when sharing the platform with a "noisy" neighbour?



BACKGROUND

Related Work & Our Contribution

Prior Work — Abeni & Faggioli (2019/2020)

- Characterized Xen & KVM real-time latency under synthetic stress
- Found QEMU priority-inversion bug in Xen HVM → unbounded latency
- Showed Credit2 scheduler introduces structural jitter
- Stack: Linux 4.x, Xen ≈4.11, older CPUs

Our Extension — Modern Stack

- Linux 6.18.35, PREEMPT_RT 6.18.35-rt5, Xen 4.17.3
- TACLe real-world benchmarks (5 programs, up to 1M loops)
- 32-scenario stressor sweep (cache / interrupts / rawsock)
- Noisy DomU study (dedicated "neighbour" VM)
- PV / PVH / HVM comparison

Key Contributions

✅ **Verify** prior findings on a decade-newer software stack

🔧 **Confirm** priority-inversion fix in Xen 4.17.3

📊 **Identify** optimal configuration for hard real-time determinism

⚠️ **Reveal** nuanced trade-offs in pinning and scheduler choice

Anchor Papers

- L. Abeni, D. Faggioli — *IEEE ISORC 2019*
- L. Abeni, D. Faggioli — *JSA 2020*
- M. Cinque et al. — *JSS 2024*

02

Experimental Setup

Hardware testbed, kernel configurations, OS-level isolation, and measurement methodology.

SETUP

Testbed & Kernel Configurations

Hardware Testbed

Component	Details
CPU	AMD (MSI board) — fixed frequency (44× ratio)
C-States	Disabled in BIOS (Global C-state Control off)
Turbo/Boost	Precision Boost Overdrive disabled
Hypervisors	Both Xen 4.17.3 & KVM on same host

Kernel Combinations Tested

4 configurations per hypervisor (Host/Dom0 × Guest/DomU kernel)

	NRT Guest / DomU	RT Guest / DomU
NRT Host / Dom0	**NRT-NRT**	**NRT-RT**
RT/LL Host / Dom0	**RT-NRT**	**RT-RT / LL-RT ← best**

KVM Host uses full PREEMPT_RT patch

XEN Dom0 uses Low-Latency kernel (PREEMPT_RT incompatible with PV Dom0)

Kernel Versions

Kernel	Version	Used in
Standard NRT	Linux 6.18.35	NRT configs
PREEMPT_RT	6.18.35-rt5	KVM host, RT guests
Low-Latency	6.18.35-rt5-ll	Xen Dom0, LL-NRT

Both hypervisors tested on the **same physical machine** to eliminate hardware variability. Fixed CPU clock eliminates P-state-induced jitter.

SETUP

OS-Level Isolation & Measurement Tools

Isolation Techniques Applied

- `isolcpus` — prevent OS from scheduling tasks on RT cores
- IRQ affinity steering — route hardware interrupts to housekeeping cores only
- vCPU-to-pCPU pinning — static core assignment, bypass dynamic scheduler
- `mlockall` — lock all pages in RAM, prevent page-fault latency spikes
- Tickless mode (`nohz_full`) — reduce timer interrupt noise on isolated cores
- RCU callback offloading to housekeeping cores (`rcu_nocbs`)

Measurement — cyclictest

- 5-minute sustained runs per configuration
- Wakeup interval: 50 µs, SCHED_FIFO priority 99
- Histogram threshold: 1000 µs — overflows count severe events
- Metrics: Min / Avg / WCET (Max) / Overflow count

TACLe Benchmarks

- 5 programs from TACLeBench v1.9 suite
- 10k–1M execution loops, SCHED_FIFO 98, mlockall, affinity=1
- Compiled with -O2, linked with libpthread + librt

Stress Injection

- stress-ng — CPU workers + memory workers
- Small noise: 2-vCPU 4 GB RAM | Big noise: 16-vCPU 16 GB RAM
- Dom0 stress: 22 CPU + 24 GB vm workers

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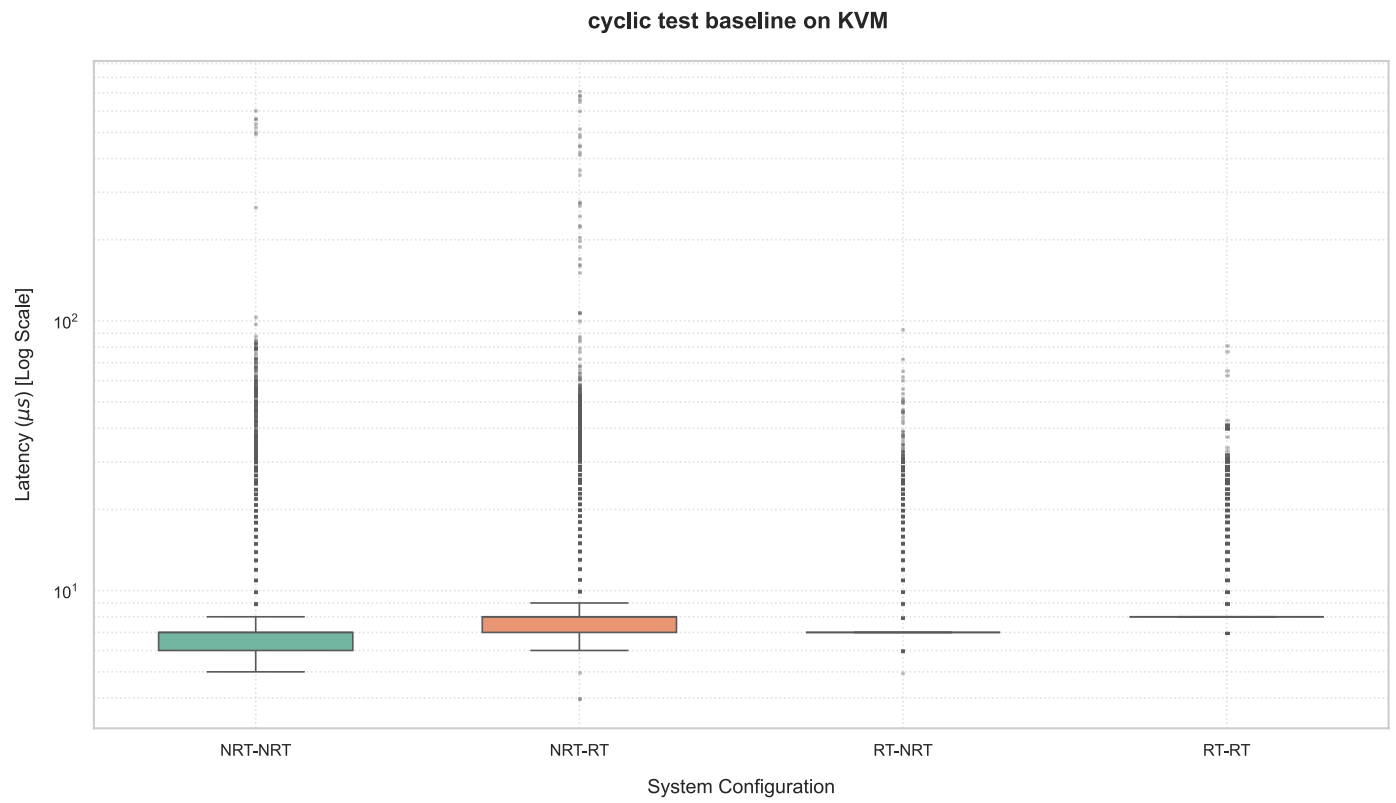
KVM
Analysis

Baseline performance, stress conditions, CPU isolation, and TACLe benchmarks on the KVM hypervisor.

KVM — BASELINE

KVM: Baseline Performance (No Stress)



Key Observations

✅ **RT-RT (Full Stack):** 81 µs WCET, 0 overflows — 92% of cycles at exactly 8 µs. Near bare-metal behaviour.

ℹ️ **RT-NRT:** Surprisingly good (93 µs) — RT host protects the guest vCPU scheduling even without a guest-side patch.

⚠️ **Determinism is end-to-end** — optimising only one side (host OR guest) is insufficient.

❌ **The NRT-RT anomaly:** An RT guest on a non-RT host achieves a WCET of 6.1 ms — worse than the fully unoptimised NRT-NRT case! A guest-side RT patch is powerless if the host cannot guarantee deterministic vCPU scheduling.

KVM — BASELINE

KVM: Baseline Performance (No Stress)

Worst-Case Latencies

Worst-Case Latencies — Baseline Configuration

Configuration	Min Latency	Average Latency ± SD	WCET (Max)	Overflows
NRT Host + NRT Guest	5 µs	7.10 ± 2.39 µs	602 µs	0
NRT Host + RT Guest	4 µs	8.55 ± 3.27 µs	6,114 µs	0
RT Host + NRT Guest	4 µs	7.22 ± 0.98 µs	93 µs	0
RT Host + RT Guest	7 µs	8.15 ± 1.15 µs	81 µs ✓	0

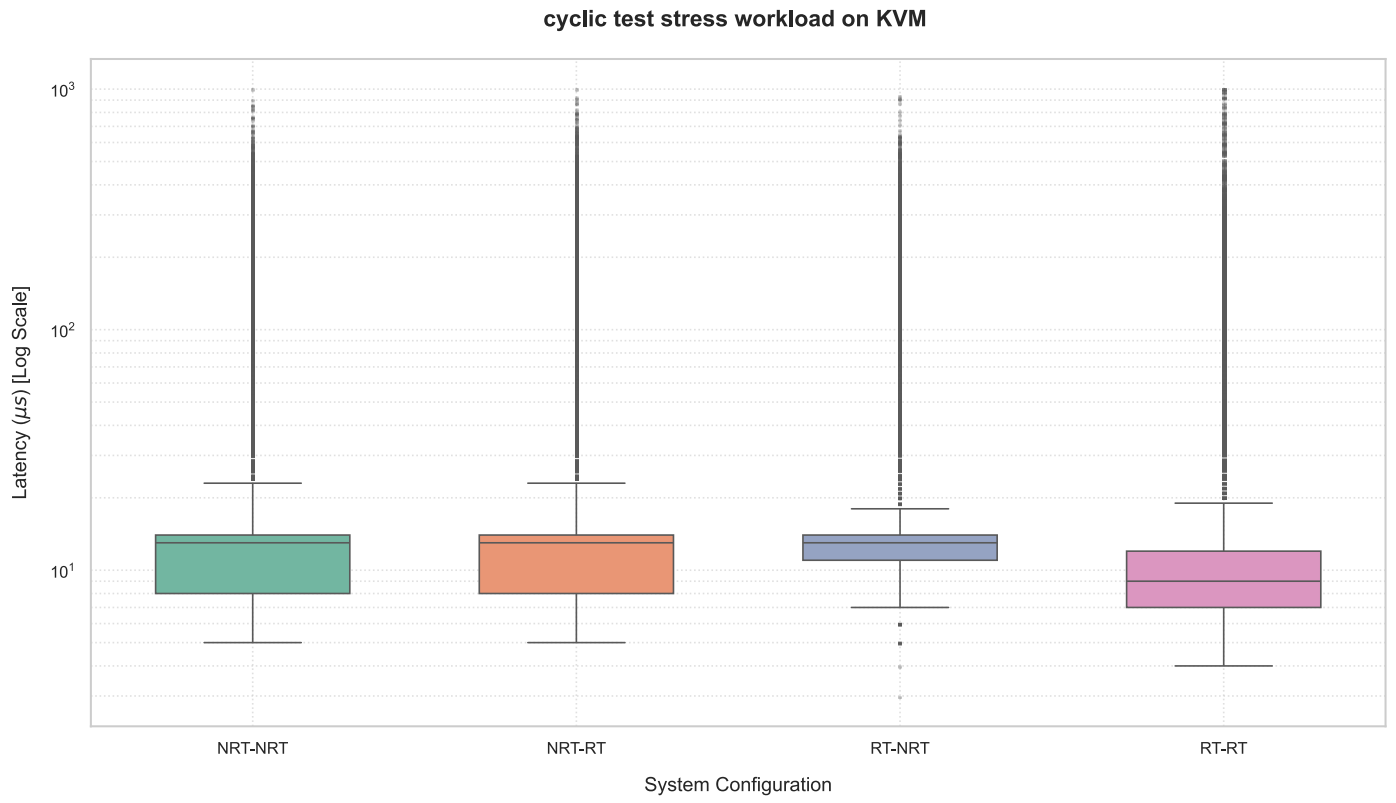
81 µs
BEST WCET (RT-RT)

6,114 µs
WORST WCET (NRT-RT)

KVM — STRESS

KVM Under Stress: Catastrophic Failure

Host-side stress-ng workload introduced (CPU + memory workers)



Key Observations

RT-RT under stress: 283 histogram overflows (latencies >1 ms). Even the best configuration fails to guarantee hard RT bounds under load.

RT-NRT percentage jump (~54,000%): The RT host was very good at baseline (93 µs) — making the stress-induced collapse appear even more dramatic proportionally.

KVM — STRESS

KVM Under Stress: Catastrophic Failure

Worst-Case Latencies

Worst-Case Latencies Under Stress

Config	Stress Average	Stress WCET	Stress Overflows	Average Δ%	WCET Δ%
NRT-NRT	13.60 ± 11.18 µs	**44,409 µs**	11	**+92%**	**+7,277%**
NRT-RT	13.69 ± 12.00 µs	**51,069 µs**	26	**+60%**	**+735%**
RT-NRT	13.33 ± 8.54 µs	**50,257 µs**	11	**+85%**	**+53,940%**
RT-RT	11.84 ± 12.79 µs	**9,829 µs**	283	**+45%**	**+12,035%**

Root Cause Analysis

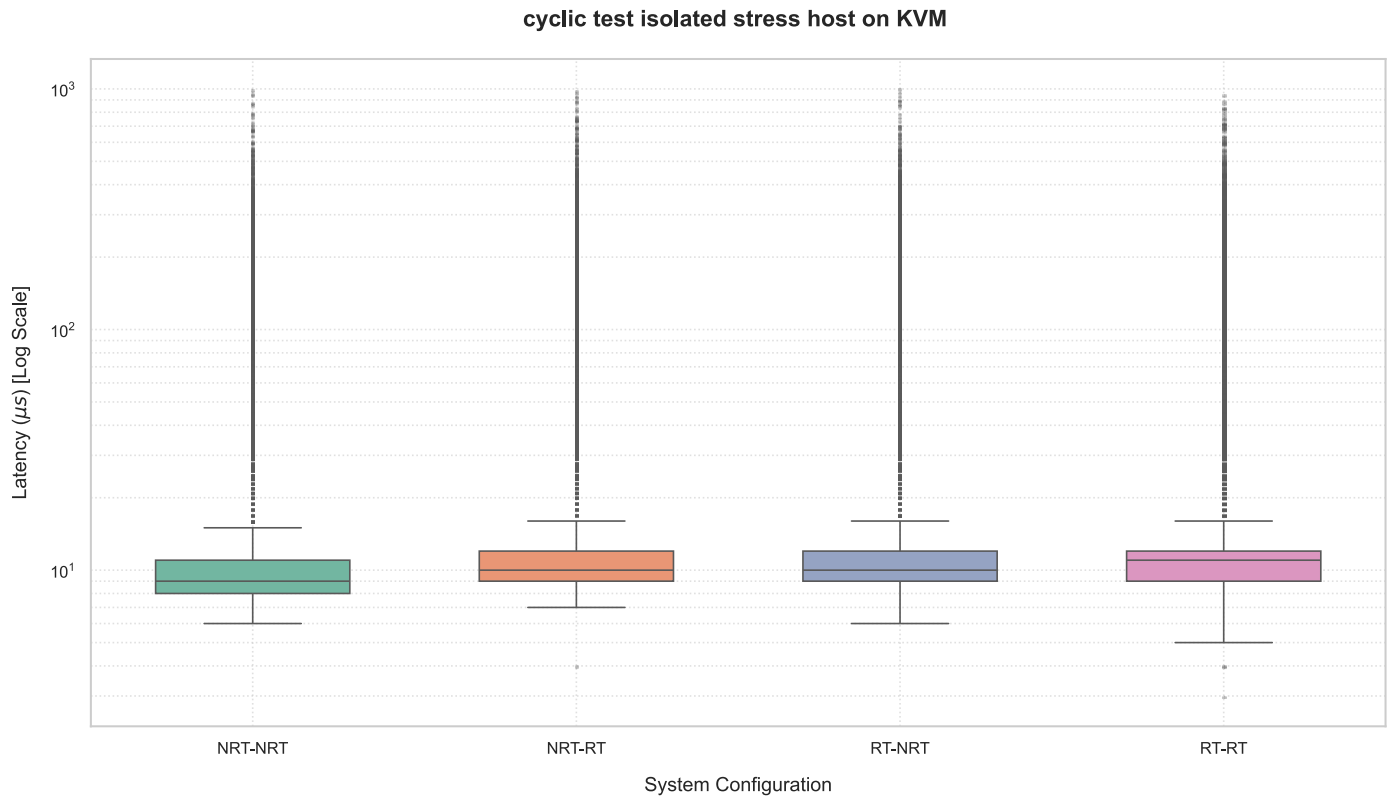
- Stress workload on host creates massive CPU contention and lock saturation
- Even PREEMPT_RT cannot prevent non-deferrable hardware interrupt storms
- Non-preemptible critical sections in kernel spinlocks starve the VM vCPU
- NRT-RT gets hit worst in absolute terms: guest RT priorities meaningless when host VCPU is starved
- RT-RT has fewer 50ms spikes but *chronic* millisecond-scale stalls (283 overflows)

❌ **Conclusion:** KVM with host-side stress cannot guarantee hard real-time bounds in any tested configuration.

KVM — STRESS

KVM: CPU Isolation — Partial Recovery

vCPU pinned to isolated cores (isolcpus + IRQ steering) while host stress runs on remaining cores



Key Observations

⚠ **Best case (RT-RT + isolated):** 1.9 ms WCET, 14 overflows. Isolation is a *massive* improvement but cannot achieve hard real-time. Residual spikes originate from hardware interrupts and non-preemptible kernel sections that cannot be prevented at the virtualization layer.

KVM — ISOLATION

KVM: CPU Isolation — Partial Recovery

Worst-Case Latencies

Worst-Case Latencies Under Stress (Isolated)

Config	Min Latency	Average Latency ± SD	WCET (Max)	Overflows
NRT-NRT	6 µs	11 µs	**2,024 µs**	19
NRT-RT	4 µs	12 µs	**2,299 µs**	11
RT-NRT	6 µs	13 µs	**1,502 µs**	16
RT-RT	3 µs	13 µs	**1,935 µs**	14

−95%

WCET REDUCTION FROM ISOLATION

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OVERFLOWS IN BEST CASE (RT-RT)

❌ **KVM hard-RT verdict:** Not suitable for applications requiring strict microsecond-level WCET bounds when co-hosting heavy workloads — even with full isolation.

KVM — TACLE BENCHMARKS

TACLeBench Suite: Real-World Workloads

Run on RT-RT (best KVM config) · 4 scenarios: Baseline / Small Noise / Big Noise / Big Noise + Pinned

matrix1
KERNEL
Raw compute, matrix operations. Cache-bound. Sub-µs avg execution time.

huff_enc
SEQUENTIAL
Data compression (325 SLOC). Single-threaded. ~16 µs avg.

lift
APPLICATION
Elevator controller (361 SLOC). Cyber-physical control loop. ~19 µs avg.

test3
STRESS TEST
Artificial WCET stress (4,235 SLOC). 10k loops. ~8,360 µs avg.

DEBIE
PARALLEL
Aerospace debris monitor (6,615 SLOC, 8 tasks). ~27,000 µs avg.

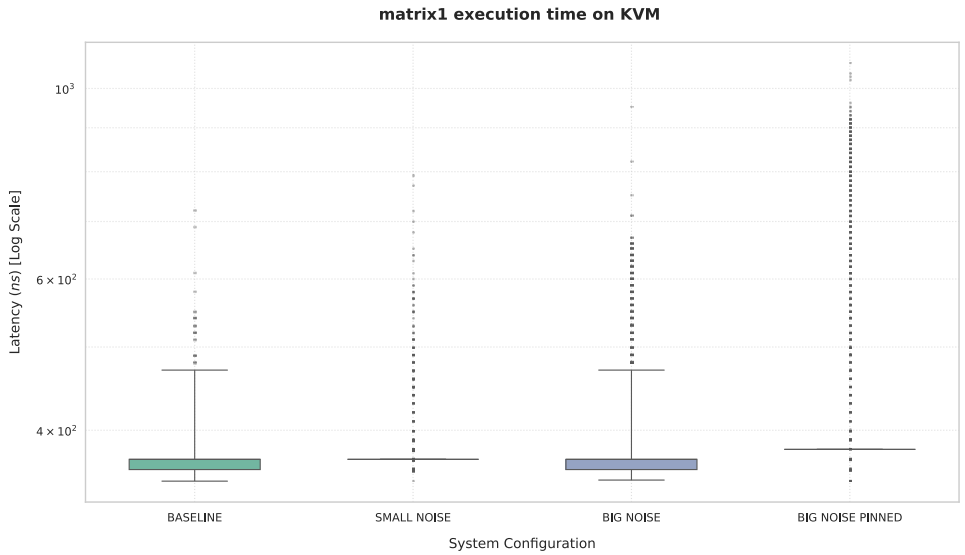
Test Scenarios (KVM: RT-RT configuration, 4 dedicated host cores)

Scenario	Noisy Guest	Pinning	Purpose
Baseline	None	No	Reference WCET without interference
Small Noise	2 vCPUs, 4 GB RAM	No	Light inter-guest interference
Big Noise	16 vCPUs, 16 GB RAM	No	Heavy inter-guest interference
Big Noise Pinned	16 vCPUs, 16 GB RAM	Yes	Effectiveness of static vCPU allocation under heavy load

Kernel scheduling isolation (isolcpus) was **not** used for TACLe — it interfered with KVM's vCPU-to-pCPU assignment. IRQ affinity + tickless mode were applied instead.

KVM — TACLE BENCHMARKS

TACLe: matrix1 — Raw Compute (Kernel Benchmark)



matrix1: Execution Time Across Scenarios

Scenario	Min (ns)	Avg ± SD (ns)	WCET (ns)	Jitter (ns)
Baseline	349	367 ± 6	**720**	371
Small Noise	350	369 ± 6	**790**	440
Big Noise	350	370 ± 20	**950**	600
Big Noise Pinned	350	384 ± 29	**1,070**	720

KVM — TACLE BENCHMARKS

TACLe: matrix1 — Analysis

Raw compute (kernel benchmark): sensitivity to noise

Analysis

- matrix1's tiny working set keeps every scenario in the sub-microsecond range: WCET spans just 0.72–1.07 µs across all four conditions
- Small Noise has only a mild effect: WCET rises from 0.72 µs to 0.79 µs (+10%) — barely perceptible at this timescale
- Big Noise is the main disturbance, but modest in absolute terms: WCET reaches 0.95 µs (+32% vs baseline) — far from the dramatic multiples seen in other benchmarks
- Pinning does **not** help here — unlike every other KVM TACLe benchmark: Big Noise Pinned reaches the highest WCET of the series (1.07 µs), +13% above Big Noise and +49% above the original baseline

⚠️ matrix1 stays sub-microsecond in every scenario — noise has a measurable but practically negligible effect. Unlike the rest of the KVM TACLe suite, **pinning does not reduce the worst case here; it slightly increases it**.

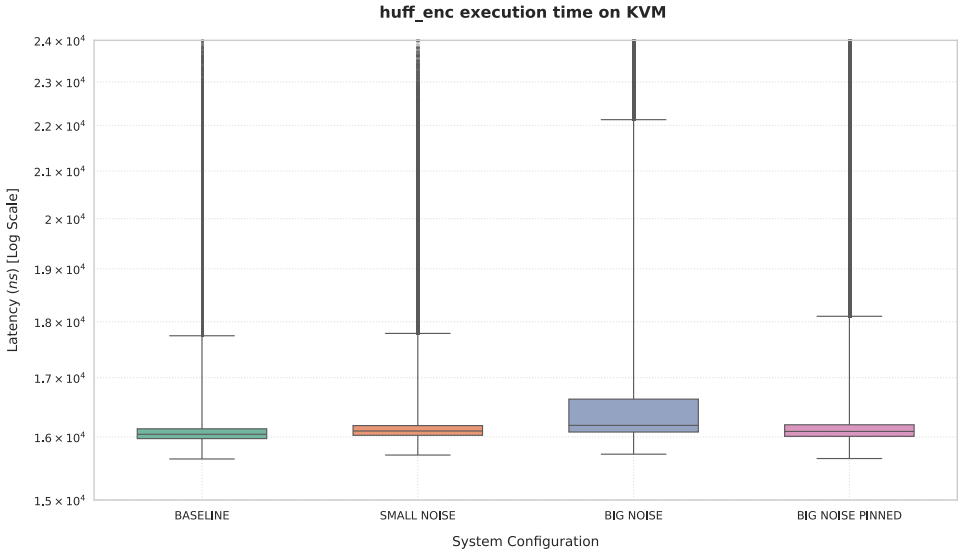
+32%
WCET INCREASE UNDER BIG NOISE

+10%
WCET INCREASE UNDER SMALL NOISE

+13%
WCET WITH PINNING (NO BENEFIT)

KVM — TACLE BENCHMARKS

TACLe: huff_enc — Data Compression (Sequential)



huff_enc: Execution Time Across Scenarios

Scenario	Min (ns)	Avg ± SD (ns)	WCET (ns)	Jitter (ns)
Baseline	15,643	16,619 ± 3,708	**65,431**	49,788
Small Noise	15,706	16,624 ± 3,207	**49,307**	33,601
Big Noise	15,720	17,306 ± 4,400	**393,480**	377,760
Big Noise Pinned	15,650	16,871 ± 3,772	**72,590**	56,940

KVM — TACLE BENCHMARKS

TACLe: huff_enc — Analysis

Data compression (sequential): sensitivity to noise

Analysis

- huff_enc's baseline (16.6 µs average, 65.4 µs WCET, 49.8 µs jitter) already carries a much wider tail than matrix1 — expected for a longer, non-cache-resident instruction mix
- Small Noise leaves the average essentially unchanged (16.6 µs) and even *lowers* the WCET to 49.3 µs (−25% vs baseline) — a statistical effect of the specific interleaving
- Big Noise is where the danger appears: the average barely moves (+4%, to 17.3 µs) but WCET explodes to 393.5 µs — a **+501%** increase, the largest relative WCET blow-up of any KVM TACLe benchmark
- Pinning strongly mitigates the tail: WCET falls back to 72.6 µs (−82% vs Big Noise), landing just 11% above the original baseline

It's the **WCET, not the average**, that reveals the danger: average latency moves only slightly (+4%) while WCET explodes by **+501%** under Big Noise. Pinning brings the tail back down to near-baseline levels (+11% residual).

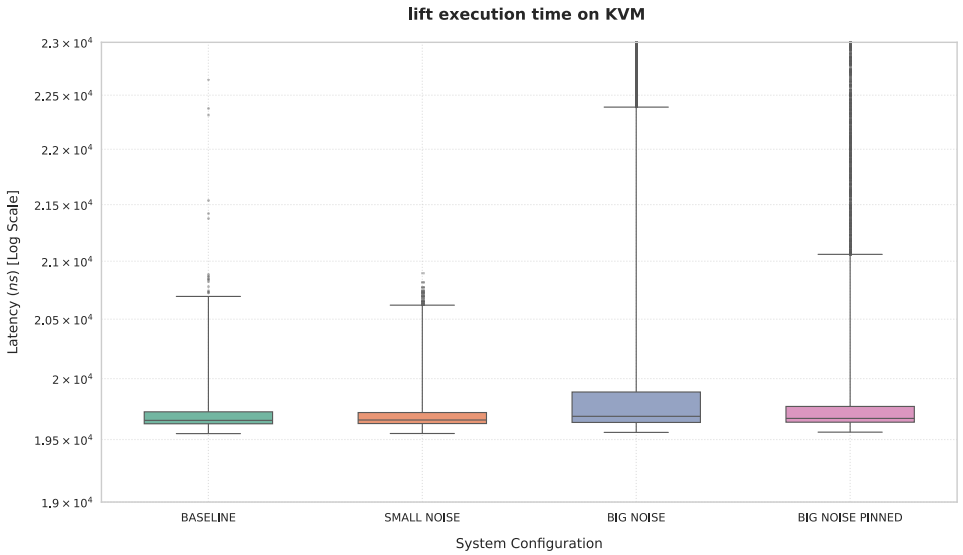
+501%
WCET INCREASE UNDER BIG NOISE

+4%
AVERAGE INCREASE UNDER BIG NOISE

−82%
WCET REDUCTION WITH PINNING

KVM — TACLE BENCHMARKS

TACLe: lift — Elevator Controller (Application)



lift: Execution Time Across Scenarios

Scenario	Min (ns)	Avg ± SD (ns)	WCET (ns)	Jitter (ns)
Baseline	19,548	20,106 ± 2,997	**57,169**	37,621
Small Noise	19,549	20,113 ± 3,034	**54,767**	35,218
Big Noise	19,558	21,218 ± 4,413	**186,424**	166,866
Big Noise Pinned	19,560	21,259 ± 5,273	**69,974**	50,414

KVM — TACLE BENCHMARKS

TACLe: lift — Analysis

Elevator controller (application): sensitivity to noise

Analysis

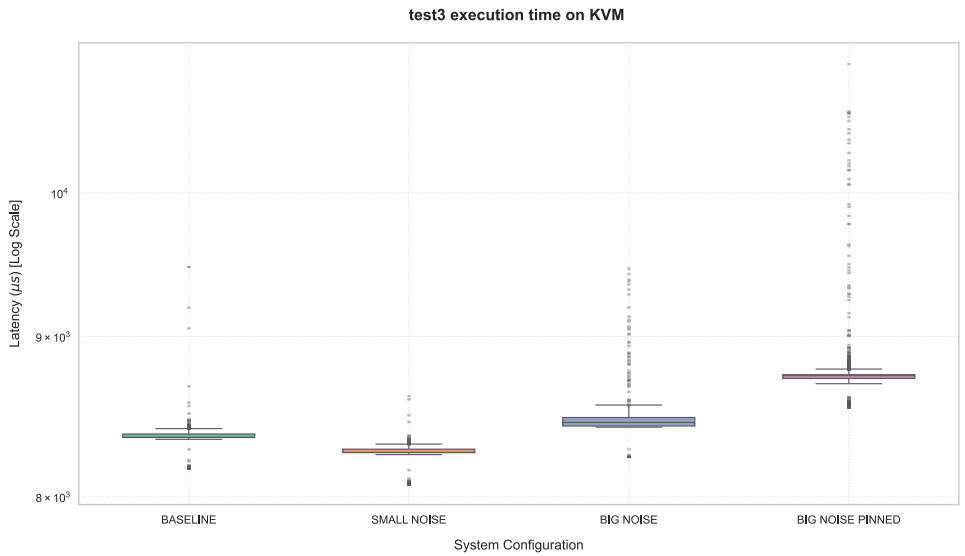
- lift's baseline (20.1 µs average, 57.2 µs WCET, 37.6 µs jitter) already shows a noticeable tail even on the unisolated scheduler
- Small Noise barely changes behaviour: average unchanged (20.1 µs), WCET even slightly lower (54.8 µs, −4% vs baseline)
- Big Noise causes a real breakdown in determinism: WCET jumps to 186.4 µs — **+226% vs baseline (3.3× baseline)** — while the average barely moves (+5.5%)
- Pinning substantially reduces the spike: WCET falls to 70.0 µs (−62% vs Big Noise), though this remains ~22% above the original baseline — a strong but incomplete recovery

Dangerous tail growth: average stays nearly flat (+5.5%) but WCET jumps to 186 µs (+226%, 3.3× baseline) under Big Noise. Pinning brings it back down to 70 µs — a strong (−62%) but not complete recovery, still ~22% above the true baseline.

3.3×	166.9 µs	−62%
WCET DEGRADATION UNDER BIG NOISE	JITTER UNDER BIG NOISE	WCET REDUCTION WITH PINNING (PARTIAL)

KVM — TACLE BENCHMARKS

TACLe: test3 — WCET Stress Test



test3: Execution Time Across Scenarios

Scenario	Min (µs)	Avg ± SD (µs)	WCET (µs)	Jitter (µs)
Baseline	8,169	8,364 ± 32	**9,471**	1,302
Small Noise	8,078	8,270 ± 25	**8,612**	534
Big Noise	8,243	8,458 ± 52	**9,458**	1,215
Big Noise Pinned	8,544	8,745 ± 87	**10,996**	2,452

KVM — TACLE BENCHMARKS

TACLe: test3 — Analysis

WCET stress test: sensitivity to noise

Analysis

- test3's baseline already carries substantial jitter (1,302 µs) — a medium-weight computational loop naturally more exposed to scheduling noise than lightweight kernels like matrix1
- Small Noise paradoxically tightens the jitter (534 µs, the lowest of the series) — a statistical effect of the specific interleaving, not a sign of added protection
- Big Noise remains close to baseline (1,215 µs jitter) — this loop's medium duration appears to average out short bursts of interference
- Pinning is counter-productive here: WCET rises to 10,996 µs and jitter more than doubles baseline (2,452 µs) — the highest of all four scenarios

Unlike every other TACLe benchmark, **pinning does not help test3** — it produces the worst WCET (10,996 µs) and worst jitter (2,452 µs) of the series. Confining a medium-weight, longer-running loop to isolated cores may **starve it of the scheduling flexibility it needs**, or concentrate it alongside residual system interrupts on that core.

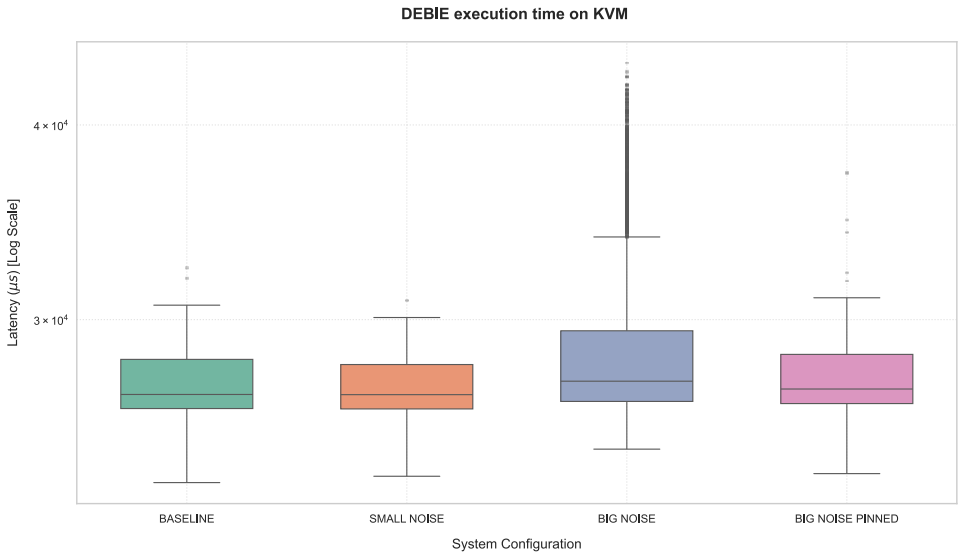
1,302 µs
BASELINE JITTER

−59%
JITTER IMPROVEMENT UNDER SMALL NOISE

+88%
JITTER DEGRADATION WITH PINNING

KVM — TACLE BENCHMARKS

TACLe: DEBIE — Aerospace Multi-Task Parallel Benchmark



DEBIE: Execution Time Across Scenarios

Scenario	Min (µs)	Avg ± SD (µs)	WCET (µs)	Jitter (µs)
Baseline	23,578	27,214 ± 1,304	**32,409**	8,831
Small Noise	23,800	27,221 ± 1,305	**30,872**	7,072
Big Noise	24,773	28,846 ± 3,814	**43,849**	19,076
Big Noise Pinned	23,893	27,444 ± 1,328	**37,284**	13,391

KVM — TACLE BENCHMARKS

TACLe: DEBIE — Analysis

Multi-task parallel workload: sensitivity to noise

Analysis

- DEBIE's 8-task parallel structure makes it inherently sensitive to preemption and synchronisation overhead
- Big Noise nearly **doubles the jitter** (8,831 → 19,076 µs)
- Small noise actually *reduces* WCET vs baseline — stochastic effect of scheduling
- Pinning helps (37,284 µs vs 43,849 µs) but DEBIE's variance cannot be eliminated — inter-task synchronisation overhead persists

⚠️ Multi-task parallel benchmarks show a fundamentally different sensitivity profile than single-threaded tasks: the average grows with noise, and pinning provides only partial mitigation.

+35%	−15%	+6%
WCET INCREASE (BIG NOISE)	WCET REDUCTION (PINNING)	AVERAGE INCREASE (BIG NOISE)

KVM — TACLE SUMMARY

KVM TACLe: Max Execution Time Summary (µs)

Benchmark	Baseline	Small Noise	Big Noise	Big Noise Effect	Big Noise Pinned	Pinning Effect
DEBIE	32,409	30,872	**43,849**	**+35%**	**37,284**	**−15%**
huff_enc	65	49	**393**	**+501%**	**73**	**−82%**
lift	57	55	**186**	**+226%**	**70**	**−62%**
matrix1	0.72	0.79	**0.95**	**+32%**	**1.07**	**+13%**
test3	9,471	8,612	**9,458**	**-0.14%**	**10,996**	**+16%**

✅ **Pinning is highly effective when interference actually drives the tail:** huff_enc WCET falls 82% (393 → 73 µs) and lift falls 62% (186 → 70 µs) under Big Noise Pinned — both land close to their own unstressed baseline.

⚠️ **Pinning doesn't help every workload:** it slightly raises WCET for the already sub-microsecond matrix1 (+13%) and clearly worsens the long-running test3 (+16%, 9,458 → 10,996 µs) — concentrating interference on one core can backfire when that wasn't the dominant source of jitter to begin with.

💡 **Core insight:** The impact of noise and the benefit of pinning are highly workload-dependent. Average execution time is a poor proxy for worst-case determinism — WCET must be measured directly.

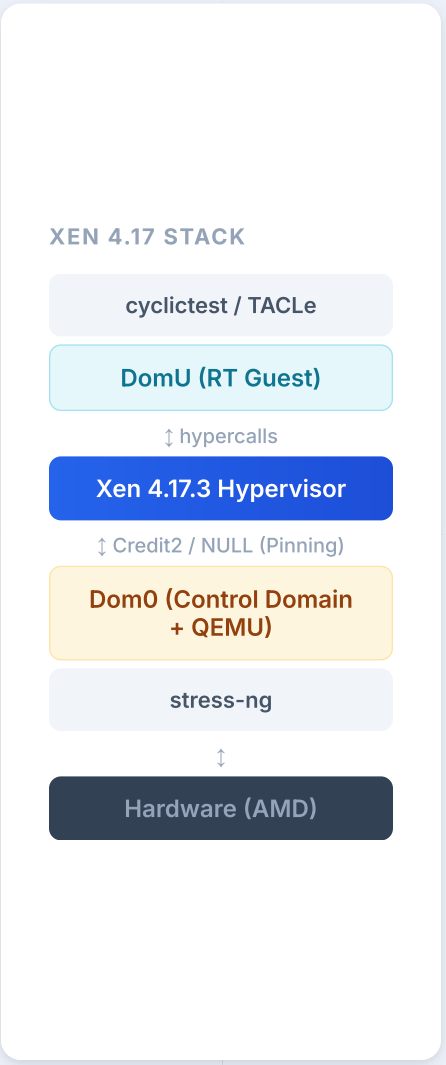
04

Xen Analysis

Scheduler comparison, priority inversion study, 32-scenario sweep, noisy DomU, TACLe benchmarks, and PV/PVH/HVM evaluation.

XEN — ARCHITECTURE

Xen: Architecture & Tested Configurations



Configurations Tested (cyclictest campaign)

Dom0 Kernel	DomU Kernel	Scheduler	Stress?
NRT	NRT / RT	Credit2	No / Yes
Low-Latency (LL)	NRT / RT	Credit2	No / Yes
NRT	NRT / RT	vCPU Pinning (NULL equiv.)	No / Yes
Low-Latency (LL)	NRT / RT	vCPU Pinning (NULL equiv.)	No / Yes

Credit2 Scheduler

- Default Xen scheduler
- Fair-share, work-conserving
- Dynamic vCPU-to-pCPU mapping
- Introduces inherent ~32 µs avg jitter

vCPU Pinning (NULL Emulation)

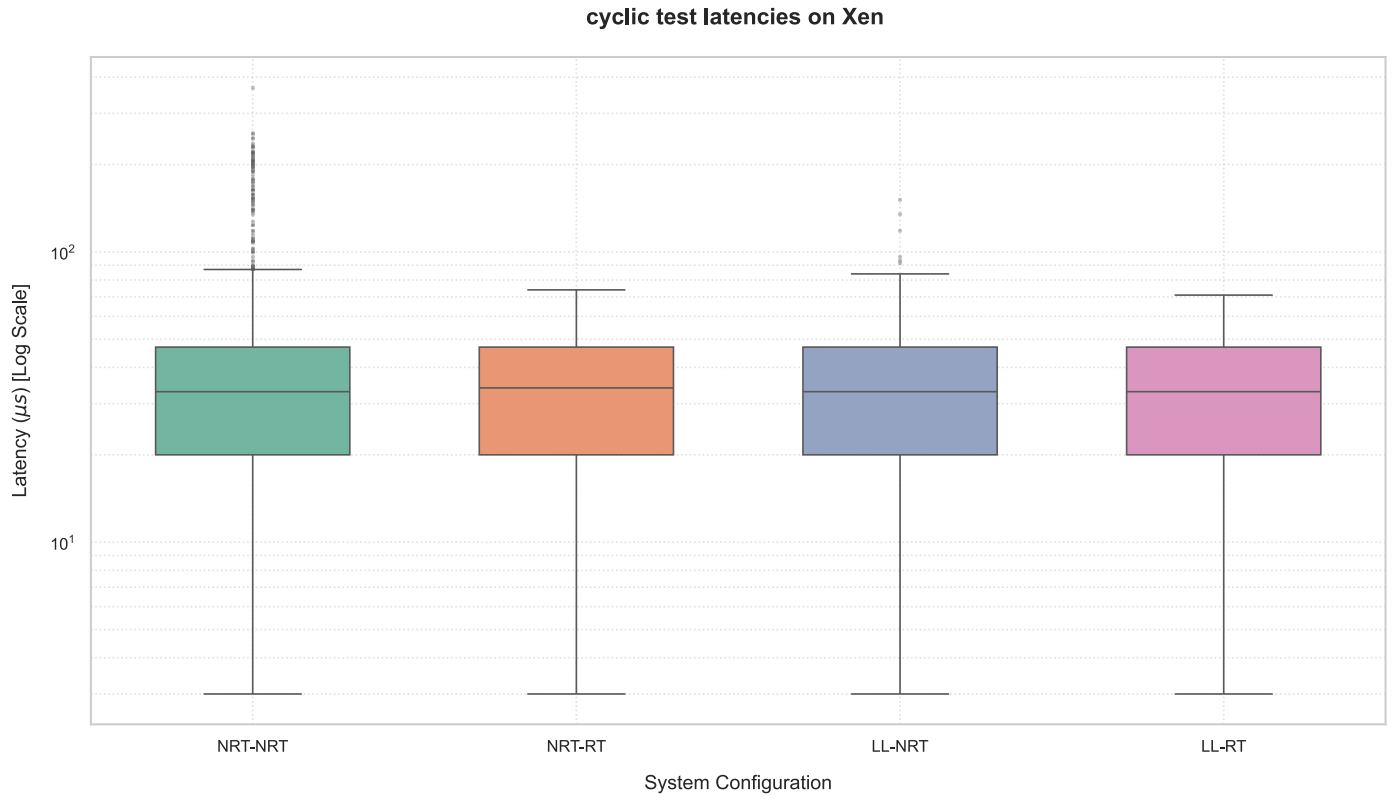
- NULL scheduler unavailable in Xen 4.17.3 (experimental)
- Static pinning: Dom0 → first N pCPUs, DomU → last M pCPUs
- Effectively bypasses Credit2 decisions

⚠️ **Key difference from KVM:** Xen Dom0 is a PV guest — PREEMPT_RT patch causes soft CPU lockups in PV mode. Dom0 uses a **Low-Latency** tuned kernel instead.

XEN — BASELINE

Xen Baseline: Credit2 Scheduler

No stress workload. All configurations, 5-minute cyclictest runs.



Key Observations

Hypervisor jitter is constant: Average latency is ~32 µs across all configurations — a structural Xen virtualization constant that Credit2 alone cannot reduce.

RT DomU dominates: NRT-RT achieves 74 µs — close to LL-RT. The RT guest successfully bounds its own critical sections.

Optimizing the DomU matters more than optimizing Dom0: a Real-Time guest alone recovers most of the achievable WCET improvement (−80%), while a Low-Latency Dom0 alone recovers less (−59%) — and combining both yields only a small additional gain (−4%) over the RT DomU alone.

XEN — BASELINE

Xen Baseline: Credit2 Scheduler

Worst-Case Latencies

Analysis

- NRT-NRT is the true baseline: with no optimization anywhere in the stack, Credit2's fair-share scheduling produces the widest WCET of the series (369 µs) — confirming that general-purpose scheduling alone cannot bound worst-case latency
- Host-side LL tuning helps, but less: a Low-Latency Dom0 alone (LL-NRT, no RT DomU) reduces WCET by 59% (369→152 µs) — meaningful, but well short of what guest-side RT patching achieves on its own
- Combining both gives the best, but diminishing, return: LL-RT reaches the series minimum (71 µs), only 4% better than NRT-RT alone — most of the achievable gain already comes from the DomU side; Dom0 tuning mainly refines an already-good result

Worst-Case Latencies — Credit2 Baseline

Configuration	Min Latency	Average Latency ± SD	WCET (Max)	Overflows
NRT-NRT	3 µs	32.87 ± 15.17 µs	369 µs	0
NRT-RT	3 µs	33.20 ± 15.12 µs	74 µs	0
LL-NRT	3 µs	32.71 ± 15.17 µs	152 µs	0
LL-RT	3 µs	32.89 ± 15.14 µs	71 µs	0

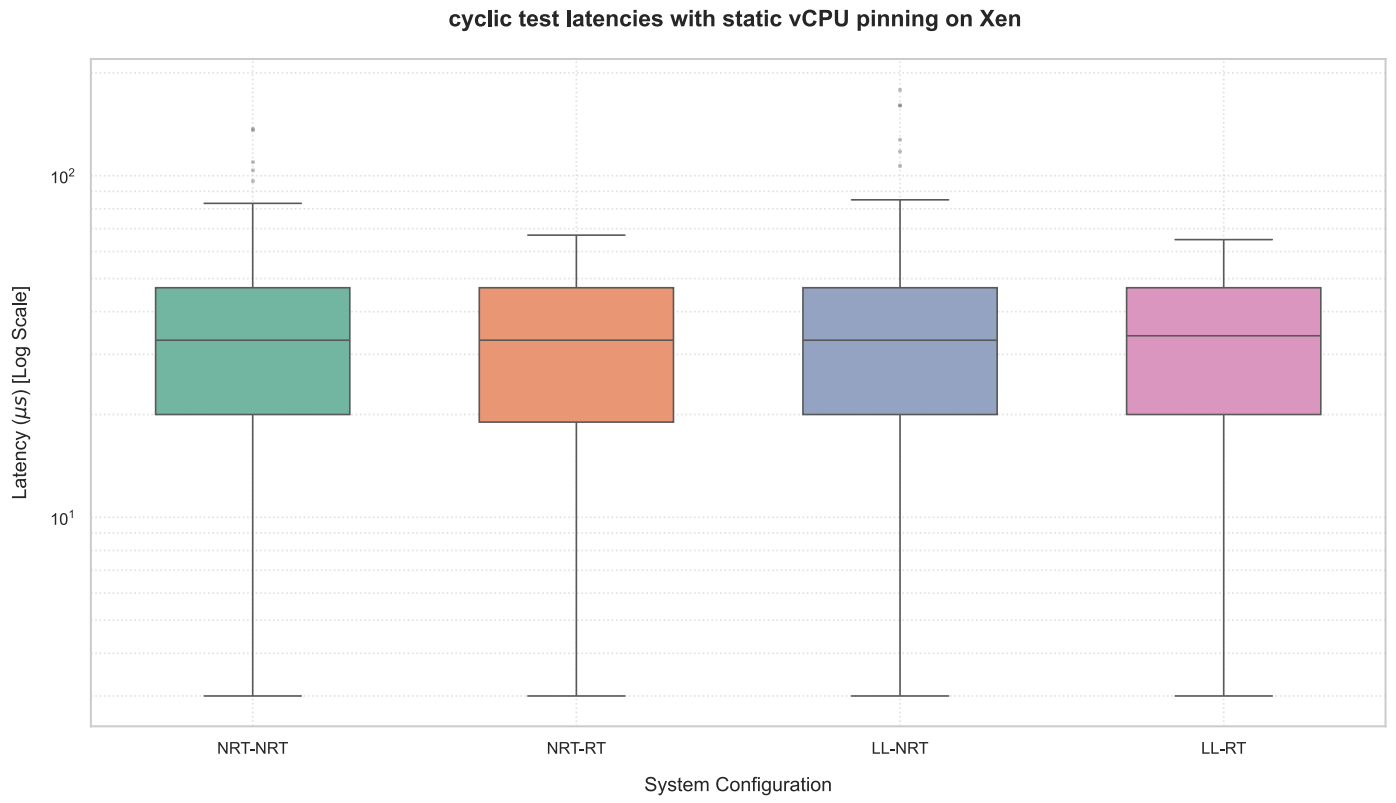
71 µs
BEST WCET (LL-RT)

369 µs
WORST WCET (NRT-NRT)

XEN — ISOLATION

Xen Baseline: vCPU Pinning

No stress workload. All configurations, 5-minute cyclictest runs.



Key Observations

⚠️ **LL-NRT pinning anomaly:** Pinning worsens WCET for NRT DomU on LL Dom0 (152→179 µs). LL Dom0 optimizations may concentrate certain interrupt patterns on guest-used cores.

✅ **LL-RT is best:** 65 µs WCET — excellent for a baseline, comparable to KVM RT-RT (81 µs).

ℹ️ **vCPU pinning under Credit2** is used here as a practical substitute for Xen's Null Scheduler, which could not be enabled in this version. Results should be read as "near-static" rather than fully offline scheduling — some residual Credit2 jitter may remain.

XEN — ISOLATION

Xen Baseline: vCPU Pinning

Worst-Case Latencies

Analysis

- Pinning is a workaround, not the intended configuration: the Null Scheduler (a purely static, offline allocator) could not be enabled — Xen 4.17.3 rejected it and CPU-pool creation failed, likely due to the scheduler's experimental status in this version. vCPU pinning under Credit2 was used as a practical emulation of the same static-allocation behaviour
- This explains the residual jitter: because Credit2 remains technically active underneath the pinning configuration, some latent scheduling overhead from Credit2's internal logic may still leak through — the pinned results approximate, but are not strictly identical to, a true offline/static scheduler
- The close match to expected values validates the approach: the pinned WCETs (65–179 µs) are close enough to the theoretical Null Scheduler behaviour described in the reference methodology to confirm pinning is a reasonable proxy, even if not a perfect one

Worst-Case Latencies — vCPU Pinning

Configuration	Min Latency	Average Latency ± SD	WCET (Max)	Overflows
NRT-NRT	3 µs	32.65 ± 15.17 µs	137 µs	0
NRT-RT	3 µs	32.59 ± 15.16 µs	67 µs	0
LL-NRT	3 µs	32.49 ± 15.19 µs	179 µs	0
LL-RT	3 µs	33.09 ± 15.18 µs	65 µs	0

65 µs

BEST WCET (LL-RT PINNED)

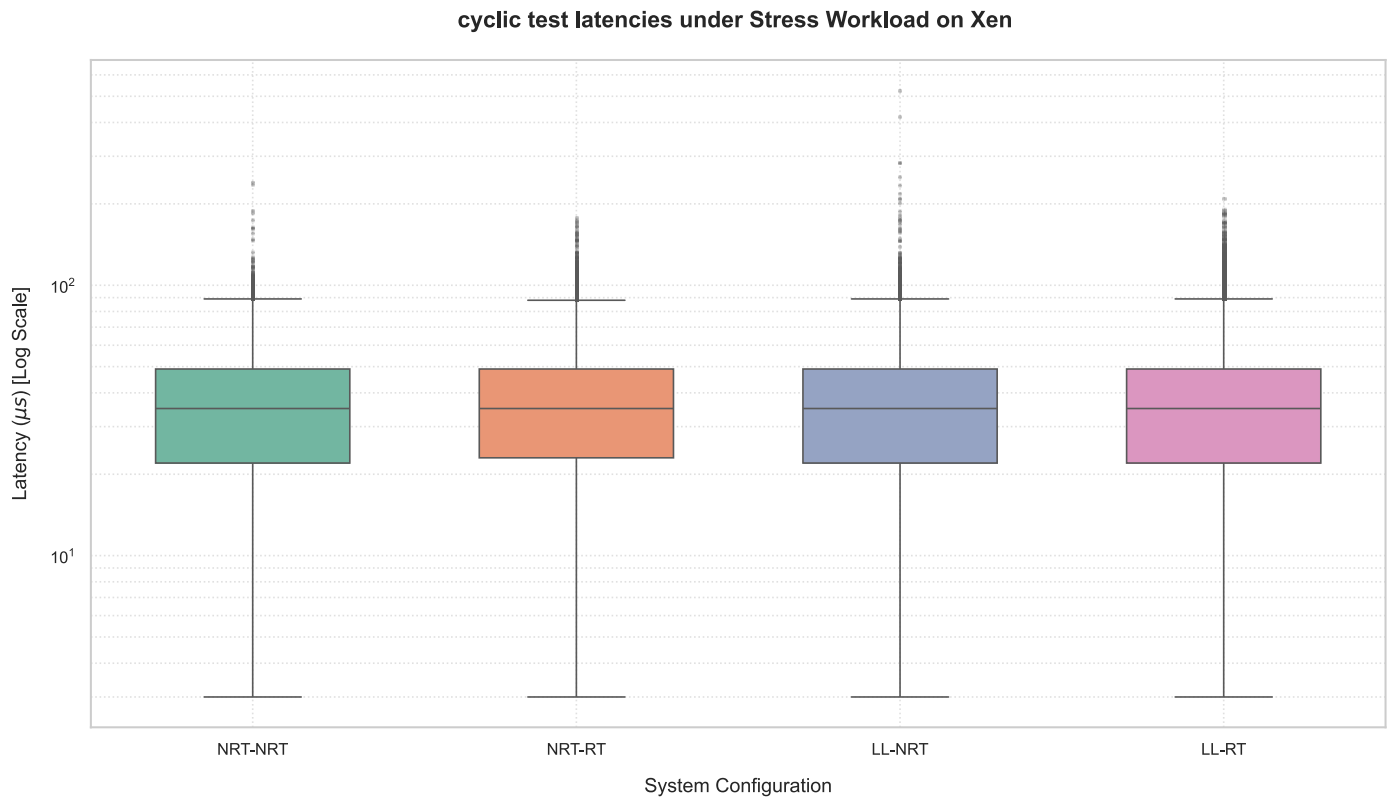
−63%

MAX IMPROVEMENT FROM UNPINNED (NRT-NRT)

XEN — STRESS

Xen Under Stress: Credit2 Scheduler

Dom0 stress workload applied (stress-ng CPU + memory)



Key Observations

- **LL-NRT Credit2 Stress:** 526 µs — worse than NRT-NRT (239 µs)! Low-latency Dom0 tuning without a corresponding RT DomU can *concentrate* interference and worsen the tail.

XEN — STRESS

Xen Under Stress: Credit2 Scheduler

Worst-Case Latencies

Analysis

- The ~35 µs average is unshakeable: even under sustained Dom0 stress-ng load, average latency stays pinned at 35 µs across all four configurations — confirming this is Xen's structural virtualization floor, not a function of load or kernel tuning
- RT DomU is the only reliable shield: whether paired with a standard or Low-Latency Dom0, a PREEMPT_RT guest consistently bounds WCET under stress (177 µs / 209 µs) — proving that guest-side real-time patching is the dominant protective factor, robust to what happens on the host side
- Optimizing the wrong side backfires: this is the clearest evidence in the dataset that Dom0-only tuning is not a substitute for DomU-side RT patching — and can be counter-productive if applied in isolation
- Ranking under stress: NRT-RT (177 µs) ≈ LL-RT (209 µs) ≪ NRT-NRT (239 µs) ≪ LL-NRT (526 µs) — the presence of an RT DomU matters roughly 3× more than the presence of an LL Dom0

Worst-Case Latencies Under Stress — Credit2

Config	Stress Average	Stress WCET	Overflows	Average Δ%	WCET Δ%
NRT-NRT	35.52 ± 15.50 µs	239 µs	0	+8%	−35%
NRT-RT	35.86 ± 15.54 µs	177 µs	0	+8%	+139%
LL-NRT	35.58 ± 15.55 µs	526 µs	0	+9%	+246%
LL-RT	35.66 ± 15.60 µs	209 µs	0	+8%	+194%

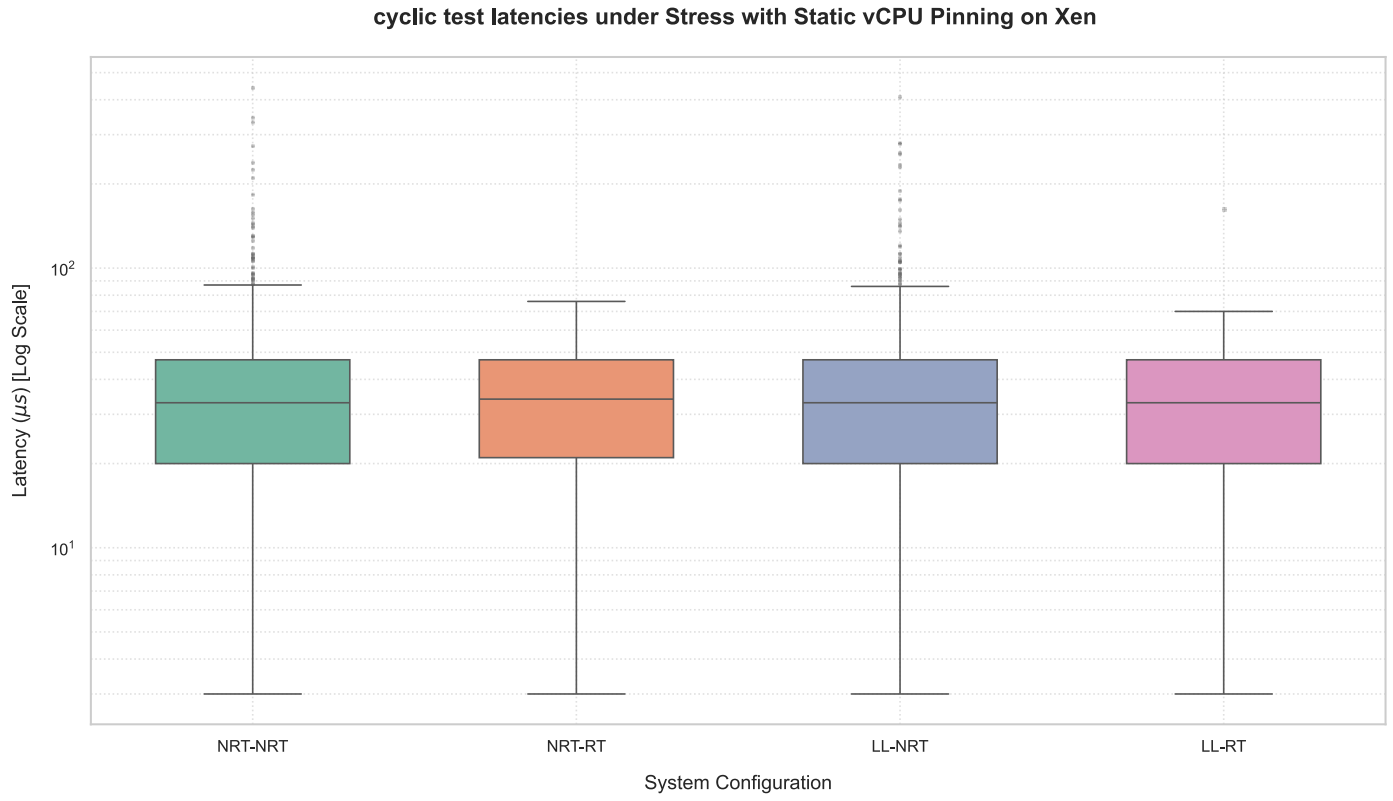
526 µs
WORST WCET (LL-NRT)

177 µs
BEST WCET (NRT-RT)

XEN — STRESS + ISOLATION

Xen Under Stress: vCPU Pinning

Dom0 stress workload applied (stress-ng CPU + memory)



Key Observations

⭐ **NRT-RT + Pinning Stress: 76 µs** — remarkably robust. Only 13% above its unstressed baseline. The RT DomU protects itself even without an LL Dom0.

💡 **Key insight:** The *combination* matters more than the individual optimization. An RT DomU is the single most protective factor. Pinning under stress without an RT DomU (NRT-NRT Pinning Stress: 443 µs) is still unreliable.

XEN — STRESS + ISOLATION

Xen Under Stress: vCPU Pinning

Worst-Case Latencies

Analysis

- RT DomU remains the strongest shield even with pinning: NRT-RT under pinning + stress caps WCET at just 76 µs — the lowest of the four pinned-stress configurations, confirming that guest-side RT patching dominates regardless of scheduling strategy (dynamic Credit2 or static pinning)
- Pinning only partially rescues the LL-NRT anomaly: 526 µs (Credit2, no pinning) improves to 410 µs with pinning — a 22% reduction, but still the worst result among all pinned-stress configurations, well above even the unoptimized NRT-NRT baseline (443 µs is close, but NRT-NRT's own pinning story is different — see below)
- Dom0 tuning without DomU tuning remains the weak link: across both scheduling strategies (Credit2 and static pinning), LL-NRT is consistently the worst or near-worst performer under stress — reinforcing that Low-Latency Dom0 tuning cannot compensate for the absence of RT optimization in the DomU, and pinning alone does not fully correct this asymmetry
- The ~32–33 µs average floor persists even with static pinning: confirming this jitter originates in Xen's structural virtualization overhead itself, not in scheduler choice — pinning bounds the tail (WCET) but does not touch the average

Worst-Case Latencies Under Stress (Pinned)

Config	Stress Average	Stress WCET	Overflows	Average Δ%	WCET Δ%
NRT-NRT	32.76 ± 15.10 µs	443 µs	0	+0.34%	+223%
NRT-RT	33.23 ± 15.09 µs	76 µs	0	+1.96%	+13%
LL-NRT	32.83 ± 15.06 µs	410 µs	0	+1.05%	+129%
LL-RT	32.73 ± 15.09 µs	163 µs	0	-1.11%	+150%

76 µs
BEST WCET (NRT-RT)

443 µs
WORST WCET (NRT-NRT)

XEN — DEVICE MODEL PRIORITY INVERSION

QEMU Priority Inversion: A Historical Bug — Now Fixed

Historical Issue (Abeni & Faggioli 2019)

- HVM DomU depends on QEMU Device Model running in Dom0
- Under Dom0 stress: QEMU (low-priority) gets preempted
- RT DomU blocked waiting for QEMU → priority inversion → **unbounded latency spikes**
- Mitigation: elevate QEMU to SCHED_FIFO priority 99

Our Investigation — Xen 4.17.3 (3 Experiments)

Experiment	QEMU Priority	Result
1 — Default QEMU	Default	**No extreme outliers. Comparable to baseline.**
2 — Max QEMU (NULL)	SCHED_FIFO 99	**No improvement. Distributions virtually identical.**
3 — Max QEMU (Credit2)	SCHED_FIFO 99	**Same result — no beneficial effect.**

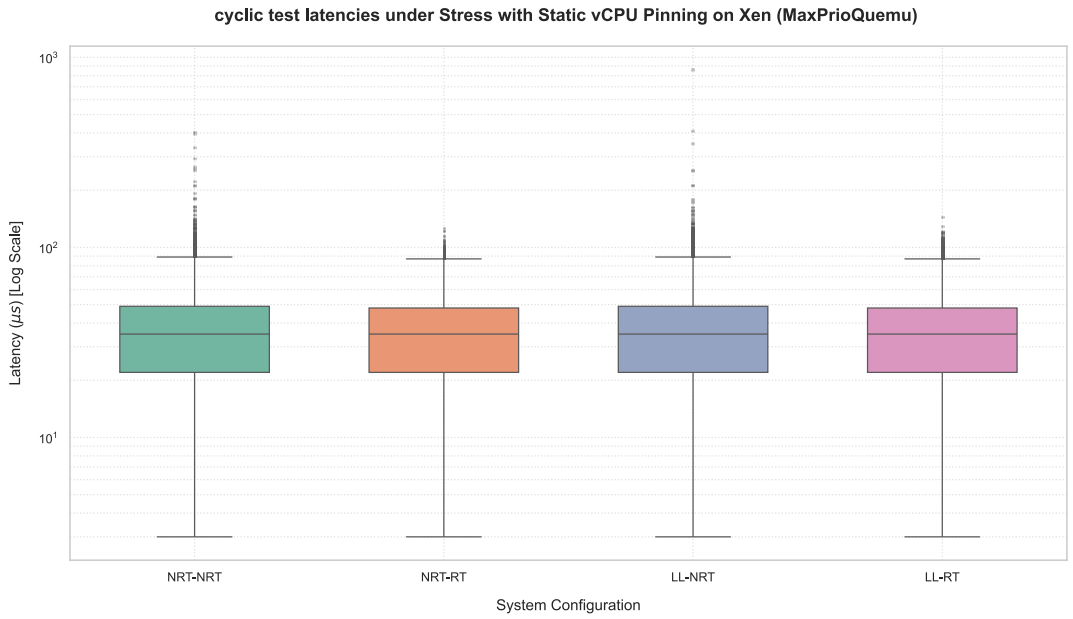
Explanation

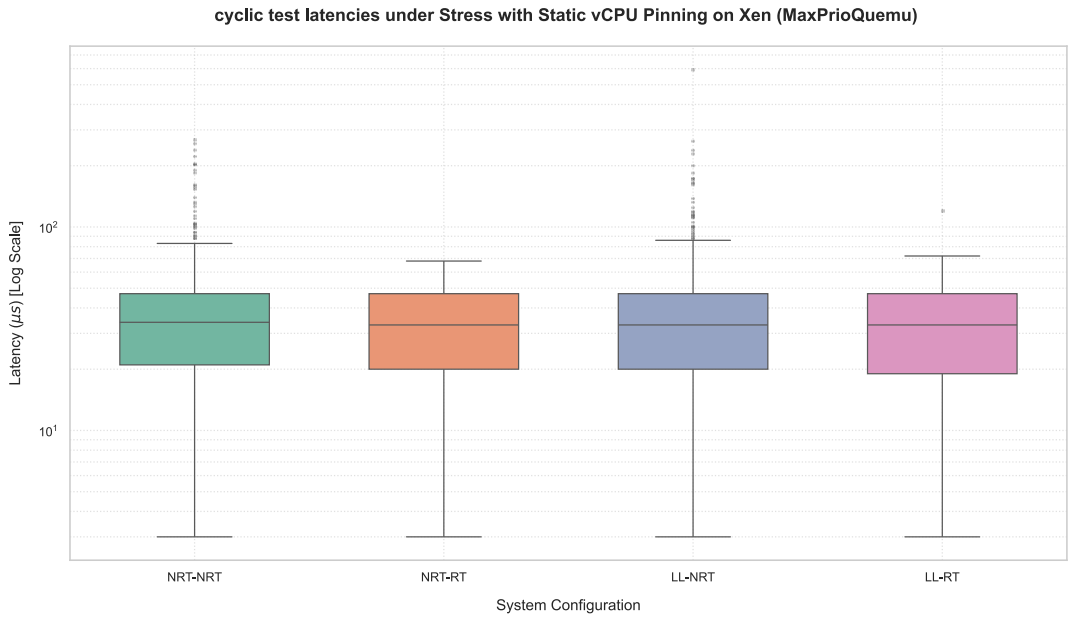
- Modern Xen HVM **decouples essential timer/ interrupt delivery** from the QEMU Device Model
- CPU-bound real-time workloads (cyclictest) exercise timer wakeups — no complex I/O requiring QEMU intervention
- DomU can maintain temporal constraints via **hardware virtualization extensions alone**
- Manually elevating QEMU priority provides no benefit and is unnecessary

✅ **Conclusion:** The priority inversion bug has been **FIXED in Xen 4.17.3**. One of the major historical concerns about Xen HVM is no longer a practical obstacle on the modern software stack.

XEN — DEVICE MODEL PRIORITY INVERSION

QEMU Priority Inversion: A Historical Bug — Now Fixed





XEN — 32-SCENARIO STRESSOR STUDY

32-Scenario Stressor Study: Summary

4 factors × combinations: Dom0 kernel (NRT/LL) · vCPU pinning (yes/no) · Guest type (nrt/rt5) · Stressor (baseline/cache/interrupts/rawsock)

Summary by Pinning Effect (all 32 scenarios)

Group	Avg Mean Latency	Max WCET	Total Overflows	Runs with Overflow
Pinned guests	**32.6 µs**	**35,984 µs***	48	**2 / 15**
Unpinned guests	35.4 µs	**61,826 µs**	179	**8 / 16**

* Single isolated anomaly — one outlier at ~36 ms, cycle 913 of 4.93M

4×
FEWER RUNS WITH OVERFLOWS (PINNED VS UNPINNED)

88
OVERFLOWS — WORST SINGLE RUN (NRT + NRT + CACHE)

Key Findings

① **vCPU pinning is the dominant factor** — almost deterministic behaviour regardless of Dom0 kernel or stressor type

② **Unpinned nrt-hvm guest is most fragile** — up to 62 ms WCET, 88 overflows under cache stressor

③ **rt5 guest systematically more stable** than nrt guest, whether pinned or not

④ **cache stressor most unpredictable** — worst single case (88 overflows) but harmless with pinning (0 overflows)

⑤ **LL vs NRT Dom0:** No clear winner on unpinned guests — interaction with guest type and stressor is more complex

Experimental Design: 4 Factors, 32 Combinations

The Four Factors

Dom0 Kernel
Standard ("NRT") vs. Low-Latency/PREEMPT_RT ("LL")

vCPU Pinning
Guests pinned to dedicated physical cores vs. unpinned (free Xen scheduling)

Guest Type
Standard Linux ("nrt") vs. RT-patched kernel ("rt5")

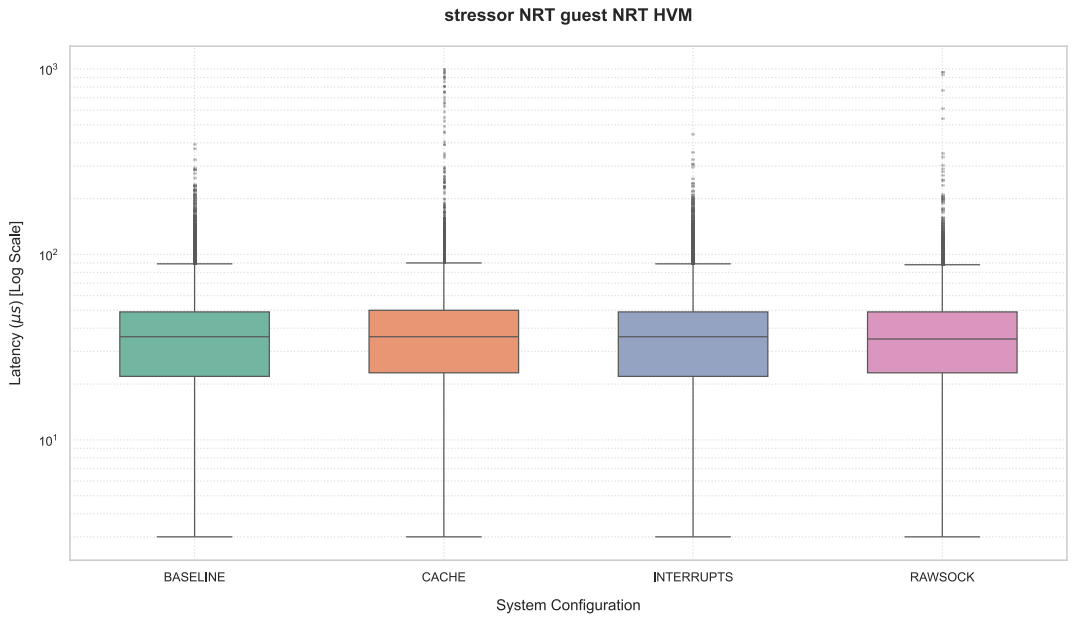
Dom0 Stressor
none (baseline) · cache · interrupts · raw sockets — on top of always-on CPU+memory load

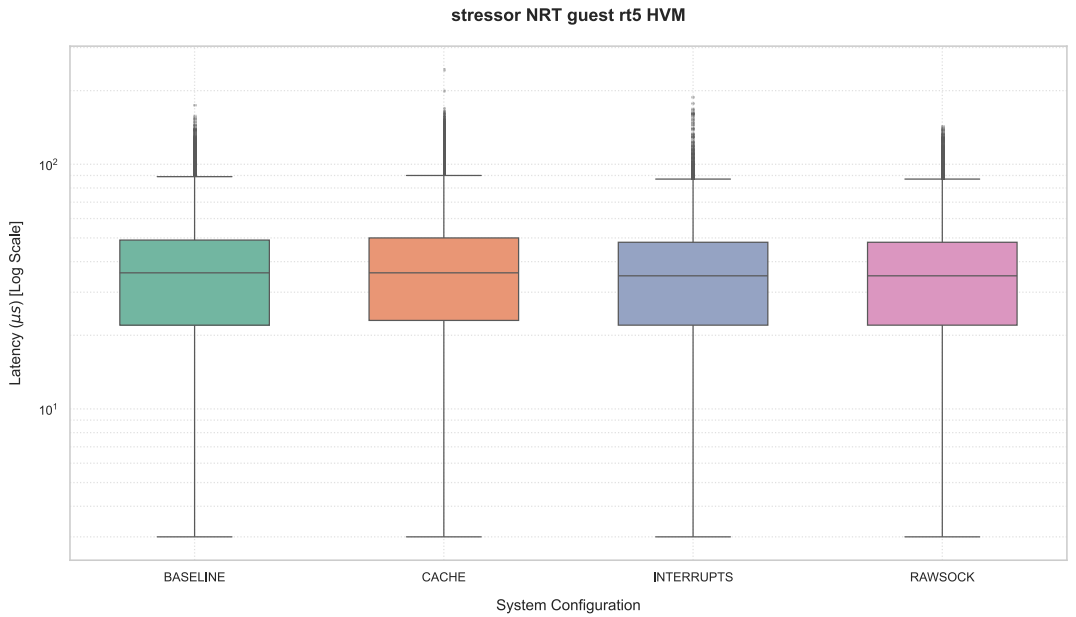
Protocol per Scenario

- Always-on stressors started on Dom0: `stress-ng --cpu 22, --vm 12 --vm-bytes 2G`, held for the guest's entire test block
- Scenario-specific sequential stressor added (none / `--cache` / `--interrupts` / `--rawsock 22`)
- DomU created, 30 s settling time, then `cyclictest` runs for 5 minutes (priority 99, 50 µs interval, 1000 µs histogram threshold)
- DomU destroyed, histogram retrieved, all stressors terminated, Dom0 caches cleared before the next scenario

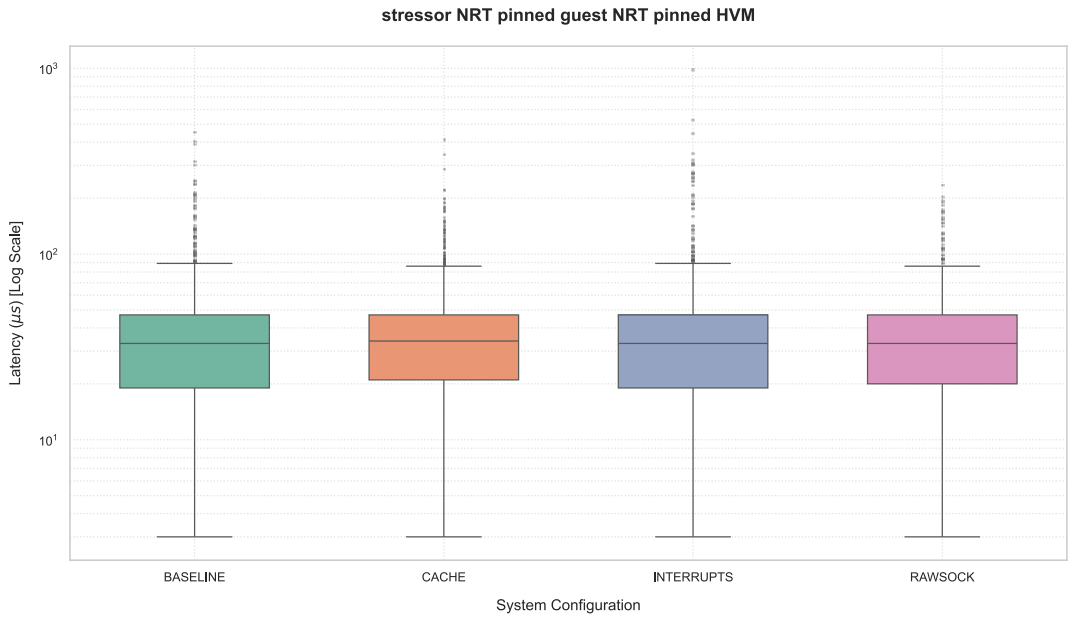
ℹ️ Each of the 32 kernel × guest × stressor combinations was executed in a **single run** (no repetitions) — isolated anomalies (see later slide) cannot be statistically distinguished from transient noise with this design.

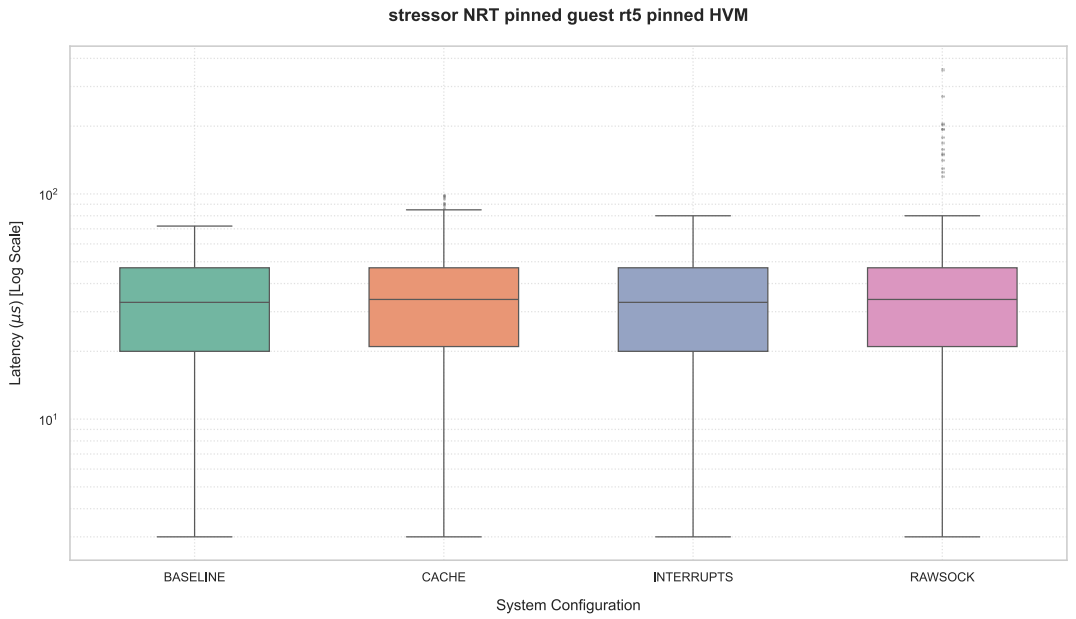
32-Scenario Stressor Study: NRT Dom0





32-Scenario Stressor Study: NRT Dom0 (Null Scheduler)





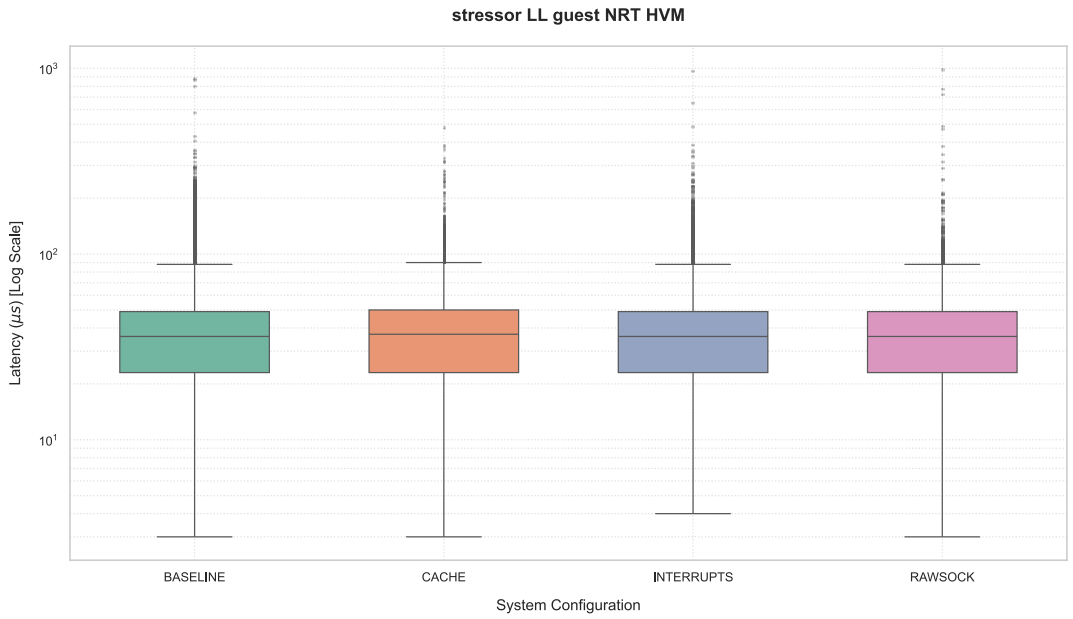
XEN — 32-SCENARIO STRESSOR STUDY

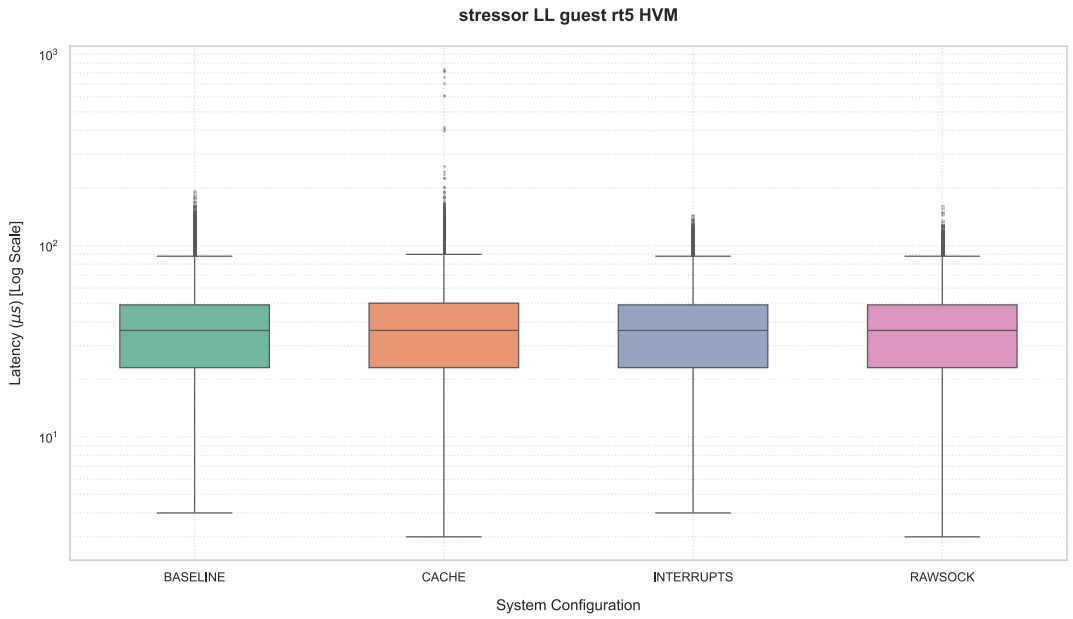
Full Results — Standard (NRT) Dom0 Kernel

16 scenarios: unpinned/pinned × nrt/rt5 guest × 4 stressors

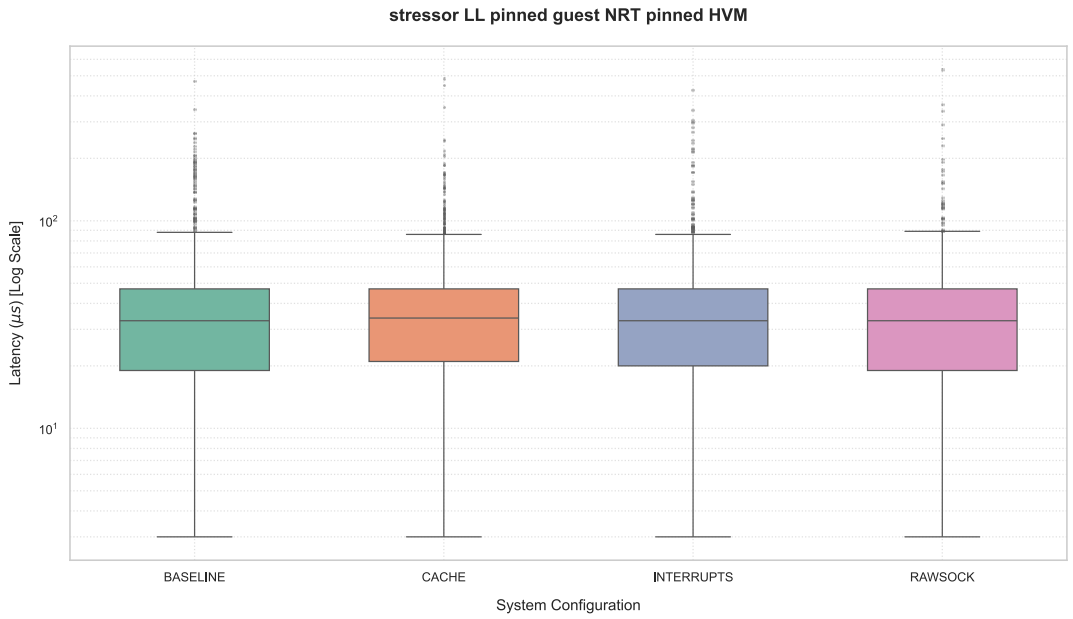
Pinning	Guest	Stressor	Avg (µs)	Max (µs)	Overflow	Note
No	nrt-hvm	baseline	35	392	0	
No	nrt-hvm	cache	36	61,611	88	*Worst case overall*
No	nrt-hvm	interrupts	35	61,826	12	
No	nrt-hvm	rawsock	35	58,344	11	
No	rt5-hvm	baseline	35	175	0	
No	rt5-hvm	cache	36	244	0	
No	rt5-hvm	interrupts	35	189	0	
No	rt5-hvm	rawsock	35	143	0	
Yes	nrt-pinned	baseline	32	450	0	
Yes	nrt-pinned	cache	33	414	0	
Yes	nrt-pinned	interrupts	32	4,225	47	*Cluster overflow, end of run*
Yes	nrt-pinned	rawsock	32	236	0	
Yes	rt5-pinned	baseline	32	72	0	
Yes	rt5-pinned	cache	34	99	0	
Yes	rt5-pinned	interrupts	33	80	0	
Yes	rt5-pinned	rawsock	33	357	0	

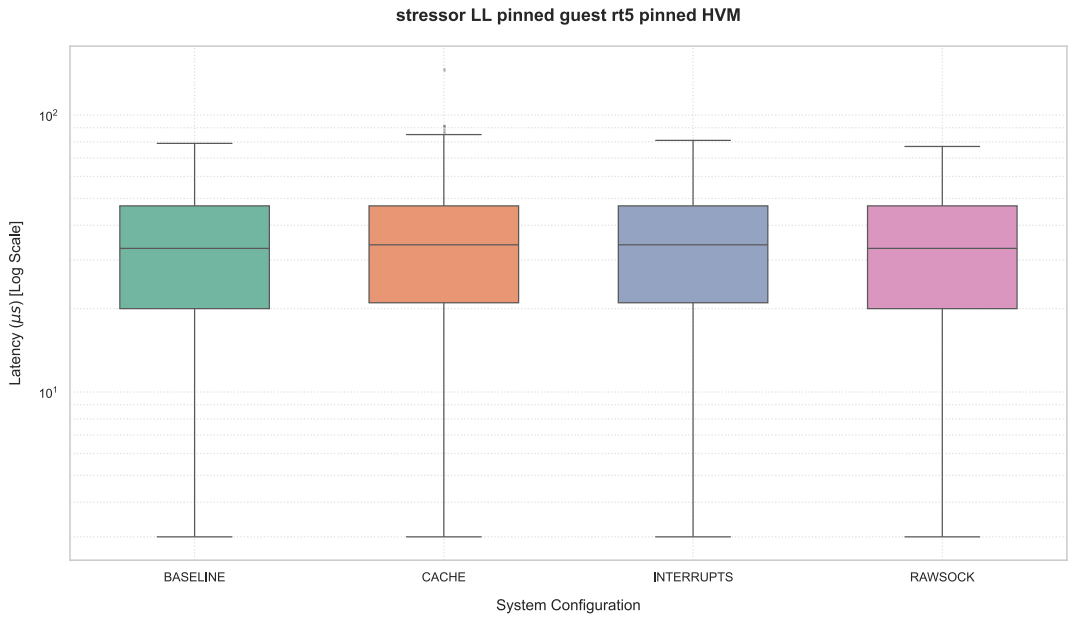
32-Scenario Stressor Study: LL Dom0





32-Scenario Stressor Study: LL Dom0 (Null Scheduler)





XEN — 32-SCENARIO STRESSOR STUDY

Full Results — Low-Latency (LL) Dom0 Kernel

16 scenarios: unpinned/pinned × nrt/rt5 guest × 4 stressors

Pinning	Guest	Stressor	Avg (µs)	Max (µs)	Overflow	Note
No	nrt-hvm	baseline	36	57,384	17	
No	nrt-hvm	cache	36	2,774	1	
No	nrt-hvm	interrupts	36	59,998	18	
No	nrt-hvm	rawsock	35	52,363	13	
No	rt5-hvm	baseline	36	192	0	
No	rt5-hvm	cache	36	2,772	19	*Isolated anomaly*
No	rt5-hvm	interrupts	35	144	0	
No	rt5-hvm	rawsock	35	162	0	
Yes	nrt-pinned	baseline	32	470	0	
Yes	nrt-pinned	cache	34	35,984	1	*Single overflow, start of run*
Yes	nrt-pinned	interrupts	32	423	0	
Yes	nrt-pinned	rawsock	32	536	0	
Yes	rt5-pinned	baseline	34	79	0	
Yes	rt5-pinned	cache	33	146	0	
Yes	rt5-pinned	interrupts	32	81	0	
Yes	rt5-pinned	rawsock	32	77	0	

XEN — 32-SCENARIO STRESSOR STUDY

Isolated Anomalies & the Cache Stressor

Three events that deviate from otherwise clean behaviour

Three Isolated Anomalies

- **LL, rt5-hvm, cache:** 19 overflows, max 2.77 ms — while the same guest under every other stressor (including cache with the NRT kernel) stays at 0 overflows
- **LL-pinned, nrt-pinned, cache:** a single overflow at 35.98 ms, occurring at cycle 913 of ~4.93M — right at the start of the run; the rest of the run is clean (secondary max ~536 µs elsewhere)
- **NRT-pinned, nrt-pinned, interrupts:** 47 overflows, but all clustered in the final ~3% of cycles — not distributed through the test

⚠ None of these three events has a counterpart in the same guest × stressor combination tested with the *other* kernel — suggesting single-run experimental noise or transient host interference rather than a reproducible kernel behaviour. With no repetitions in this design, the two hypotheses cannot be distinguished with certainty.

Why "Cache" Is the Wildcard

- Produces the single worst result of the entire study (88 overflows, NRT kernel)…
- …yet is completely harmless in the same guest/kernel combination once pinning is active (0 overflows)
- Is the only stressor involved in **all three** isolated anomalies above
- Compared to interrupts and rawsock — which impact the unpinned nrt-hvm guest more uniformly and predictably — cache shows the highest behavioural variance of any tested condition

88 → 0
OVERFLOWS: NO PINNING → PINNING (CACHE, NRT)

3 / 3
ANOMALIES INVOLVE THE CACHE STRESSOR

XEN — 32-SCENARIO STRESSOR STUDY

Final Considerations

- **vCPU pinning influences latency quality more than any other factor** — almost zero overflows across all pinned configurations, regardless of Dom0 kernel or stressor
- **The rt5-hvm (RT-patched) guest outperforms nrt-hvm in every tested condition**, pinning aside — RT guest patching is a robust, independent source of protection
- **The unpinned nrt-hvm guest is unreliable under load** (except in the plain baseline scenario), with peaks up to ~62 ms regardless of the Dom0 kernel used
- **The Low-Latency (LL) Dom0 kernel does not guarantee a systematic improvement** over the standard (NRT) kernel in this dataset — direct comparisons show crossed results depending on the specific guest/stressor combination
- **The cache stressor generates the greatest variability** of any tested condition — including both the absolute worst case of the study and all three isolated anomalies

✅ **Takeaway:** for Xen HVM guests under Dom0-side stress, *vCPU isolation* and *guest-side RT patching* are the two levers that reliably improve determinism — Dom0 kernel choice alone is not a substitute for either.

XEN — NOISY DOMU STUDY

Noisy DomU Study: Mixed-Criticality Scenario Setup

A permanently running "noisy neighbour" VM — the most realistic mixed-criticality scenario

Experimental Design

Noisy DomU (always-on)
stress-ng: 16 CPU workers + 16 × 1 GB memory workers · no timeout · continuous load

Regular DomU (under measurement)
cyclictest · 5 min · FIFO-99 · 50 µs interval · histogram 1000 µs

Dom0 (4 vCPUs)
Control domain — no additional stress from Dom0 itself

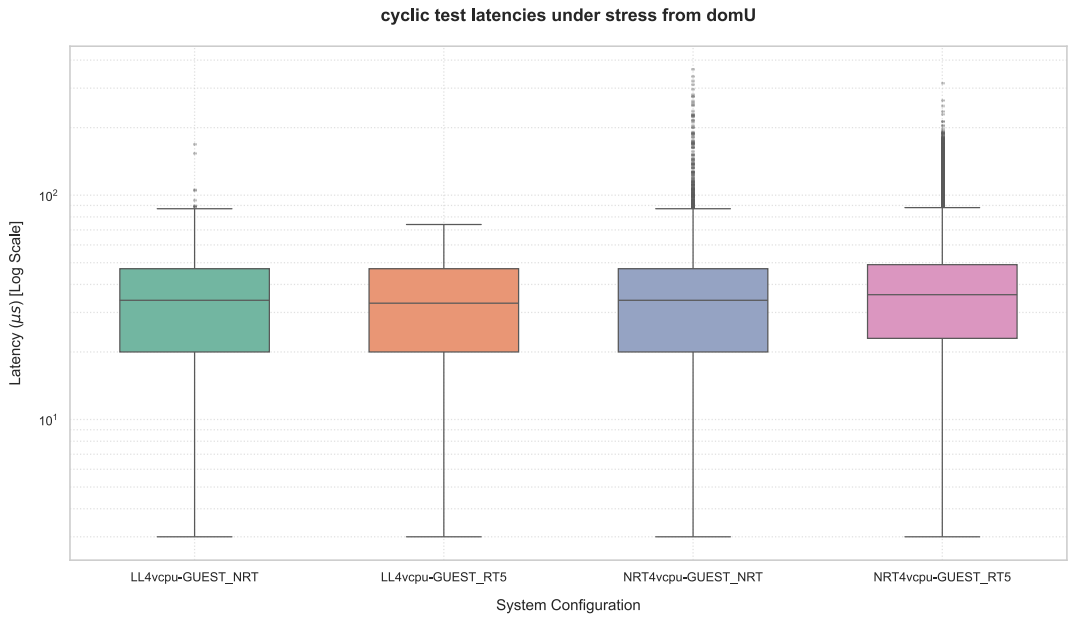
Orchestrated end-to-end by a bash script: reboots Dom0 onto the target kernel, launches the noisy DomU, then runs each regular DomU in turn — clearing Dom0 caches between runs. Each combination executed once (no repetitions).

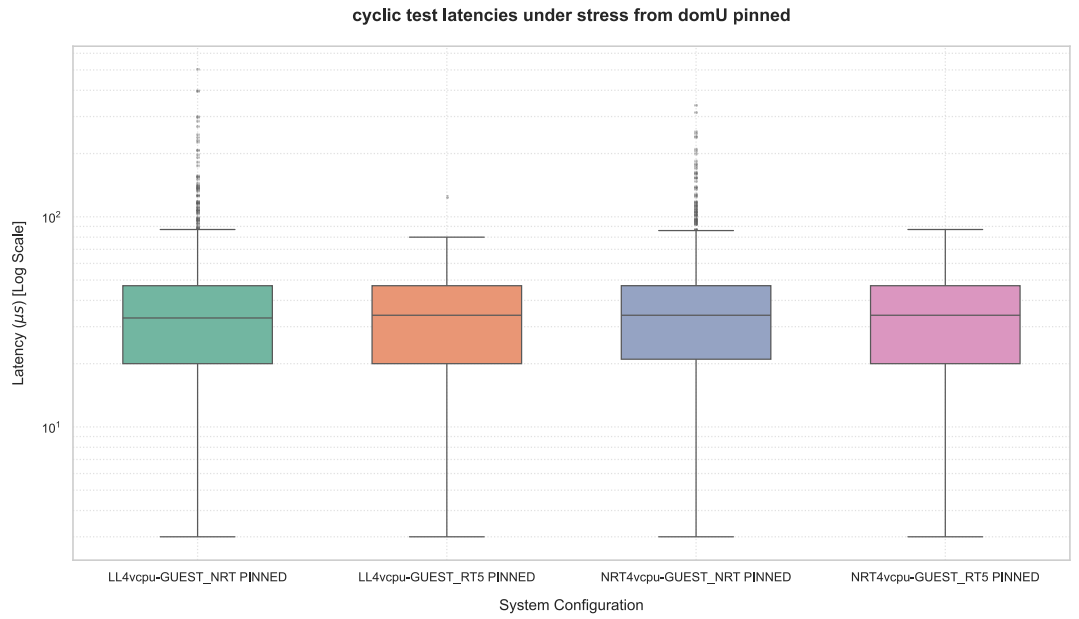
8 Test Configurations

Dom0 Kernel	Pinning	DomU Kernel
NRT-4vcpu	No	nrt-hvm (non-RT)
	No	rt5-hvm (RT)
	Yes	nrt-pinned-hvm
	Yes	rt5-pinned-hvm
LL-4vcpu (rt5-ll)	No	nrt-hvm (non-RT)
	No	rt5-hvm (RT)
	Yes	nrt-pinned-hvm
	Yes	rt5-pinned-hvm

XEN — NOISY DOMU STUDY

Noisy DomU: Results





XEN — NOISY DOMU STUDY

Noisy DomU: Results

Results — 8 Configurations

Dom0	Pinning	DomU	Avg	WCET (µs)	Overflows
NRT	No	nrt-hvm	33 µs	62,566	4
NRT	No	rt5-hvm	36 µs	312	0
NRT	Yes	nrt-hvm	33 µs	336	0
NRT	Yes	rt5-hvm	33 µs	87	0
LL	No	nrt-hvm	32 µs	167	0
LL	No	rt5-hvm	32 µs	74	0
LL	Yes	nrt-hvm	33 µs	502	0
LL	Yes	rt5-hvm	33 µs	123	0

XEN — NOISY DOMU STUDY

Isolating the Pinning & DomU-Kernel Effects

Minimum latency: 3 µs across all 8 tests

A stable and consistent value regardless of kernel, pinning, or DomU configuration — indicating that the "idle" path (no interference) is not influenced by the noisy neighbour. Only the *tail* of the distribution is affected.

🎯 This is the closest approximation to a **real mixed-criticality deployment**: a high-criticality guest running alongside a permanently active, resource-intensive low-criticality VM.

vCPU Pinning: Opposite Effects Depending on Dom0 Kernel

- **On NRT Dom0, pinning is transformative:** it eliminates the catastrophic outlier entirely — 62,566→336 µs (non-RT DomU), 312→87 µs (RT DomU) — and zeroes out all overflows
- **On LL Dom0, pinning is mildly counter-productive:** it slightly *increases* the maximum — 167→502 µs (non-RT DomU), 74→123 µs (RT DomU) — though both remain orders of magnitude below the NRT critical threshold

💡 Pinning is a powerful countermeasure **only in the absence** of a low-latency Dom0 kernel. Once Dom0 is already tuned, pinning adds no further benefit and may add slight overhead by concentrating interference onto a dedicated core.

RT DomU Kernel: Consistent Benefit in All 4 Dom0 Configs

Dom0 Config	non-RT DomU	RT DomU	Reduction
NRT-4vcpu	62,566 µs	312 µs	−99.5%
NRT-4vcpu-pinned	336 µs	87 µs	−74%
LL-4vcpu	167 µs	74 µs	−56%
LL-4vcpu-pinned	502 µs	123 µs	−75%

✅ **Overlapping benefits:** any LL-4vcpu configuration, pinned or not, stays within a few hundred µs regardless of DomU kernel. It's the total absence of *both* protections that produces the campaign's sole out-of-scale result.

XEN — NOISY DOMU STUDY

Noisy DomU: Conclusions

1 critical case out of 8: NRT + no pinning + non-RT DomU → 62.5 ms WCET, 4 overflows (at cycles 8,323 / 20,698 / 24,840 / 37,984). The total absence of both protections is the only truly dangerous combination.

Conclusions

- **7 out of 8 configurations stay under 1 ms** WCET — the noisy neighbour risk is manageable with modest mitigations
- **vCPU pinning** (on NRT Dom0): eliminates 62.5 ms → 336 µs (nrt DomU) and 312 → 87 µs (rt5 DomU)
- **LL kernel on Dom0** inherently guarantees stable behaviour regardless of pinning
- **RT DomU** systematically reduces tail latency vs non-RT twin under identical Dom0 conditions
- On LL Dom0: pinning slightly *increases* WCET (167→502 µs for nrt) — concentrated interference on a dedicated core

XEN — TACLE BENCHMARKS

Xen TACLe: Configuration & Scenario Design

Run on LL Dom0 + RT DomU (Xen's optimal configuration) — 5 benchmarks × 4 scenarios

matrix1
KERNEL
Raw compute. Cache-bound. 1M loops.

huff_enc
SEQUENTIAL
Data compression. 1M loops. ~16 µs avg.

lift
APPLICATION
Elevator controller. 1M loops. ~24 µs avg.

test3
STRESS TEST
WCET stress. 10k loops. ~8,300 µs avg.

DEBIE
PARALLEL
Aerospace multi-task. 10k loops. ~27,000 µs avg.

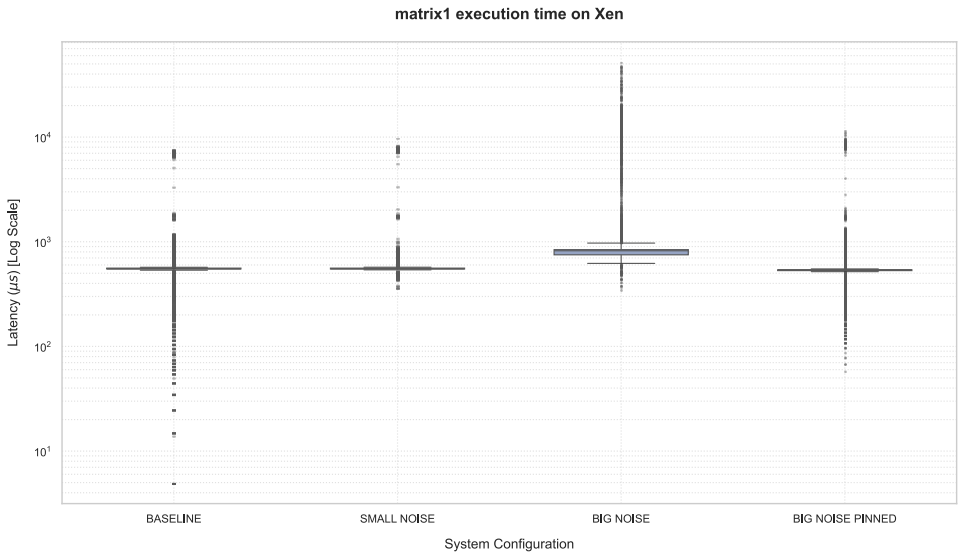
Test Scenarios (Xen: LL Dom0 + RT DomU · 4 dedicated Dom0 vCPUs)

Scenario	Noisy DomU	Pinning	Purpose
Baseline	None	No	Reference WCET without interference
Small Noise	2 vCPUs, 4 GB RAM	No	Light inter-guest interference
Big Noise	16 vCPUs, 16 GB RAM	No	Heavy inter-guest interference (stress-ng CPU+VM)
Big Noise Pinned	16 vCPUs, 16 GB RAM	Yes	Effectiveness of static vCPU allocation under heavy load

🔄 **Parallel structure with KVM TACLe:** Same 5 benchmarks, same 4 scenarios. KVM used RT-RT (KVM's best); Xen uses LL-RT (Xen's best). Direct comparison is meaningful.

XEN — TACLE BENCHMARKS

Xen TACLe: matrix1 — Raw Compute (Kernel Benchmark)



matrix1: Execution Time Across Scenarios

Scenario	Min (µs)	Avg (µs)	WCET (µs)	Jitter (µs)
Baseline	0	556 ± 154	**7,470**	7,470
Small Noise	359	557 ± 162	**9,550**	9,191
Big Noise	351	812 ± 437	**51,259**	50,908
Big Noise Pinned	58	545 ± 180	**11,321**	11,263

XEN — TACLE BENCHMARKS

Xen TACLe: matrix1 — Analysis

Raw compute (kernel benchmark): sensitivity to noise

Analysis

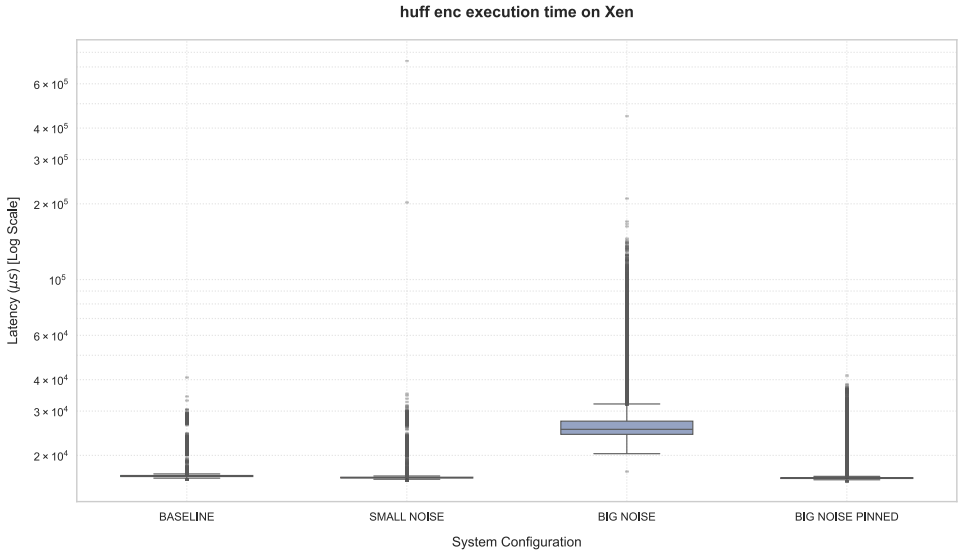
- Average latency stays under 1 µs in every scenario (0.56–0.81 µs) — the task remains essentially cache-resident throughout
- Small Noise already produces a mild disturbance on Xen: WCET rises from 7.5 µs to 9.6 µs (+28%) — unlike on KVM, where Small Noise leaves matrix1's WCET almost untouched
- Big Noise is far more disruptive on Xen than on KVM for this benchmark: WCET reaches 51.3 µs — a **6.9× increase** over baseline (vs. KVM's much milder +32% / 1.3×)
- Pinning recovers most, but not all, of the damage: WCET drops to 11.3 µs (−78% vs Big Noise), though this is still +52% above the original baseline

⚠ Unlike on KVM, matrix1 on Xen shows real sensitivity to noise — Big Noise drives WCET up nearly **7×** — and here pinning genuinely helps (−78%), even though it doesn't fully return the task to its cache-resident baseline (+52% residual).

6.9×	+52%	−78%
WCET INCREASE (BIG NOISE)	RESIDUAL WCET VS BASELINE (AFTER PINNING)	WCET REDUCTION (PINNING)

XEN — TACLE BENCHMARKS

Xen TACLe: huff_enc — Data Compression (Sequential)



huff_enc: Execution Time Across Scenarios

Scenario	Min (ns)	Avg (ns)	WCET (ns)	Jitter (ns)
Baseline	16,050	16,756 ± 1,304	**40,959**	24,909
Small Noise	15,939	16,533 ± 1,568	**740,560**	724,601
Big Noise	17,330	26,606 ± 5,118	**445,304**	427,974
Big Noise Pinned	15,879	16,568 ± 1,794	**41,370**	25,491

XEN — TACLE BENCHMARKS

Xen TACLe: huff_enc — Analysis

Data compression (sequential): sensitivity to noise

Analysis

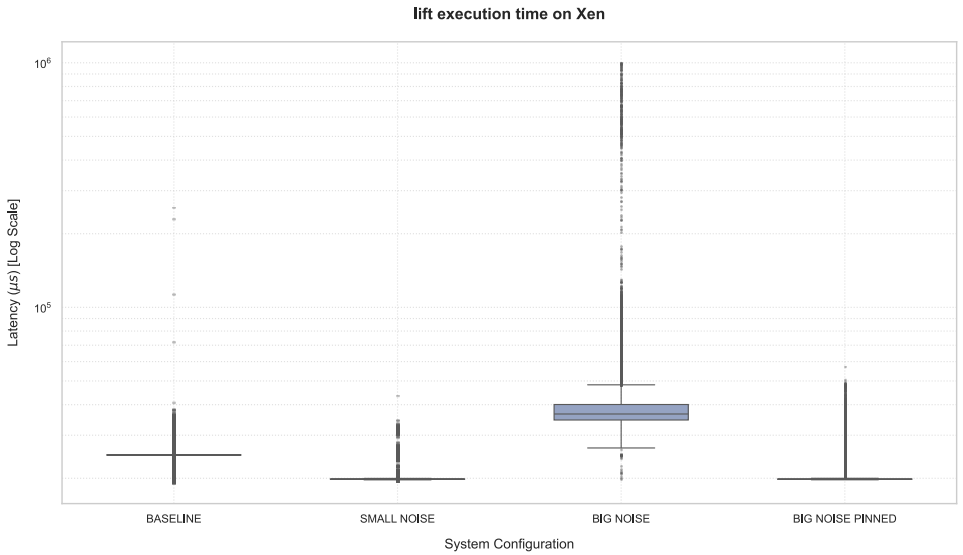
- Xen's huff_enc baseline (41.0 µs WCET, 24.9 µs jitter) is noticeably tighter than KVM's (65.4 µs) for the same benchmark
- Small Noise, surprisingly, produces the single worst WCET of the entire Xen huff_enc campaign: **740.6 µs — 18× the baseline** — a confirmed, log-verified tail-latency event from one rare preemption, occurring under a lighter noisy neighbour than "Big Noise" itself
- Big Noise also causes severe disruption: WCET rises to 445.3 µs (**+987% vs baseline, ~10.9× baseline**), while the average only rises +59% (16.8→26.6 µs)
- Pinning (tested under Big Noise) restores near-baseline behaviour: WCET falls to 41.4 µs (−91% vs Big Noise), essentially matching the unstressed baseline

Xen's biggest huff_enc surprise isn't Big Noise — it's **Small Noise**: a single rare preemption event pushes WCET to 740.6 µs, 18× baseline and even higher than the Big Noise case (445.3 µs). Pinning, tested under Big Noise, fully neutralises the disturbance (41.4 µs).

+987%	18×	−91%
WCET INCREASE (BIG NOISE)	SMALL NOISE WCET VS BASELINE (ANOMALY)	WCET REDUCTION (PINNING)

XEN — TACLE BENCHMARKS

Xen TACLe: lift — Elevator Controller (Application)



lift: Execution Time Across Scenarios

Scenario	Min (ns)	Avg (ns)	WCET (ns)	Jitter (ns)
Baseline	19,205	24,464 ± 2,225	253,587	234,382
Small Noise	19,639	20,106 ± 1,479	43,321	23,682
Big Noise	19,960	38,136 ± 11,983	1,518,396	1,498,436
Big Noise Pinned	19,720	20,254 ± 2,140	57,119	37,399

XEN — TACLE BENCHMARKS

Xen TACLe: lift — Analysis

Elevator controller (application): sensitivity to noise

Analysis

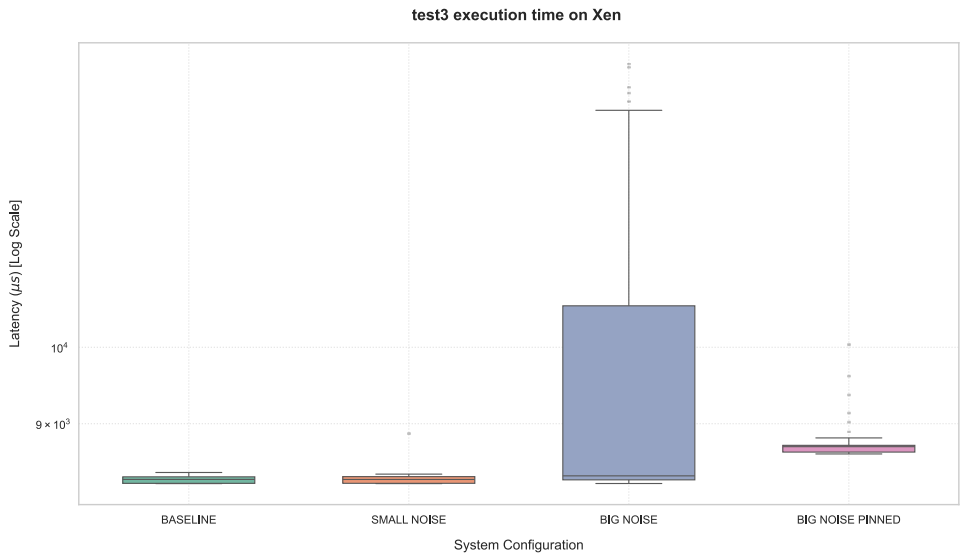
- Xen's lift baseline (254 µs WCET, 234 µs jitter) starts out much wider than KVM's (57 µs) — likely a single-run artifact given the small sample size
- Small Noise brings the WCET down sharply to 43 µs — well below even the baseline, consistent with the baseline being an outlier run rather than noise "helping"
- Big Noise causes the most severe degradation of any TACLe benchmark on Xen: WCET reaches **1,518 µs (~1.5 ms) — +499% vs baseline (6× baseline)**, with the average nearly doubling (24.5→38.1 µs, +56%)
- Pinning fully recovers determinism: WCET drops to 57.1 µs under Big Noise + Pinning — a **−96% reduction**, close to the Small-Noise figure
- **Vs. KVM:** lift's Big-Noise WCET on KVM reaches 186 µs (3.3× its own baseline); on Xen it reaches ~1,518 µs (6× its own baseline) — on this benchmark, **Xen's relative degradation under heavy noise is larger than KVM's**, though Xen's pinned recovery (57 µs) still lands below KVM's pinned result (70 µs)

Unlike matrix1, lift is **not** a case where Xen outperforms KVM under heavy noise: Big-Noise WCET reaches ~1.5 ms on Xen (6× baseline) vs. 186 µs on KVM (3.3× baseline). Pinning brings both back down to a similar, low tens-of-µs range (Xen 57 µs vs KVM 70 µs).

+499%	6×	−96%
WCET INCREASE (BIG NOISE)	BIG NOISE WCET VS BASELINE	WCET REDUCTION (PINNING)

XEN — TACLE BENCHMARKS

Xen TACLe: test3 — WCET Stress Test



test3: Execution Time Across Scenarios

Scenario	Min (µs)	Avg (µs)	WCET (µs)	Jitter (µs)
Baseline	8,286	8,328 ± 38	**8,414**	128
Small Noise	8,286	8,331 ± 40	**8,885**	599
Big Noise	8,287	9,183 ± 1317	**14,786**	6,499
Big Noise Pinned	8632	8,701 ± 44	**10,037**	1,405

XEN — TACLE BENCHMARKS

Xen TACLe: test3 — Analysis

WCET stress test: sensitivity to noise

Analysis

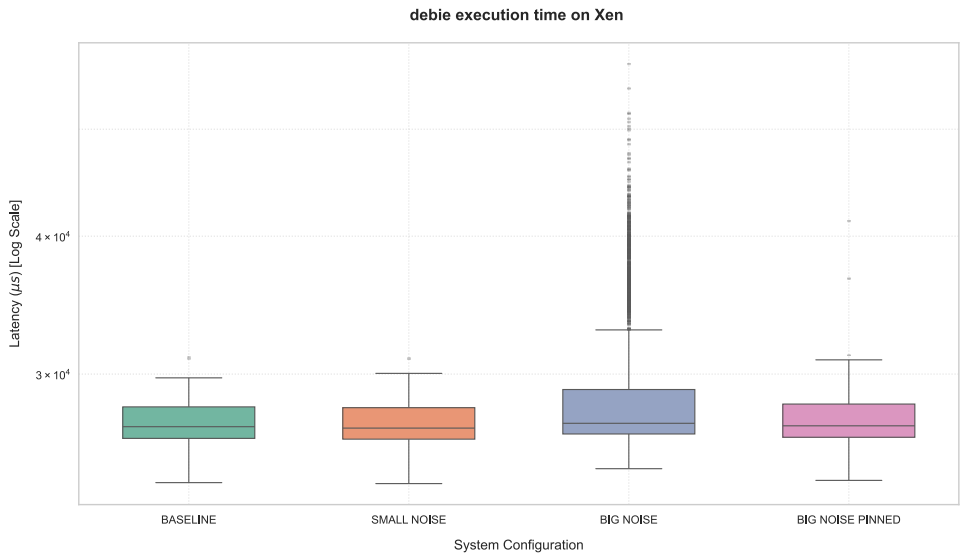
- Xen's baseline jitter (128 µs) is the tightest of any condition in this series — a well-behaved, medium-weight loop when undisturbed
- Small Noise nearly quintuples the jitter (128→599 µs) despite barely moving the average — even a light neighbor measurably lengthens the tail for this longer-running task
- Big Noise is far more severe: WCET reaches 14,786 µs (**+76%**), jitter balloons to 6,499 µs (**~51× baseline**) — the worst relative degradation of any TACLe benchmark on Xen
- Pinning recovers most, but not all: WCET drops to 10,037 µs (**−32%** from Big Noise), still well above baseline — unlike the near-full recovery seen in lighter benchmarks
- Unlike KVM's test3 anomaly (where pinning made things worse), Xen's pinning goes in the expected direction — but the recovery is partial, not complete

ℹ️ test3 is the least resilient TACLe benchmark to Dom0 noise on Xen: even after pinning, WCET remains ~1,600 µs above baseline — a residual footprint pinning alone cannot fully eliminate for this longer-running compute loop.

51×	+76%	−32%
JITTER EXPLOSION (BIG NOISE VS BASELINE)	WCET INCREASE (BIG NOISE)	WCET REDUCTION (PINNING) — PARTIAL ONLY

XEN — TACLE BENCHMARKS

Xen TACLe: DEBIE — Aerospace Multi-Task Parallel Benchmark



DEBIE: Execution Latency Across Scenarios

Scenario	Min (µs)	Avg ± SD (µs)	WCET (µs)	Jitter (µs)
Baseline	23,921	27,176 ± 1,309	31,048	7,127
Small Noise	23,873	27,118 ± 1312	31,022	7,149
Big Noise	24,627	28,197 ± 3,278	57,314	32,687
Big Noise Pinned	24,034	27,274 ± 1328	41,266	17,232

XEN — TACLE BENCHMARKS

Xen TACLe: DEBIE — Analysis

Multi-task parallel workload: sensitivity to noise

Analysis

- Noise expands the tail massively (+85% WCET) while barely touching the average (+4%) — same pattern as KVM
- Small noise has virtually *no effect* (31,022 vs 31,048 µs) — DEBIE's inherent variance absorbs it
- Pinning reduces WCET (28%) but cannot eliminate the multi-task synchronisation overhead
- Xen baseline WCET (31,048 µs) is slightly better than KVM (32,409 µs) on this parallel workload

⚠️ DEBIE's variance is fundamentally driven by its multi-task inter-task coordination, not purely by hypervisor scheduling. Pinning alone is insufficient — task-level isolation strategies are needed for such workloads.

+85%	+4%	−28%
WCET INCREASE (BIG NOISE)	AVERAGE INCREASE (BIG NOISE)	WCET WITH PINNING

XEN — TACLE SUMMARY

Xen TACLe: Summary & Pinning Trade-offs under Dom0 Stress

Max Latency Summary (µs) — Noisy DomU scenarios

Benchmark	Baseline	Small Noise	Big Noise	Big Noise Pinned	Pinning Effect
DEBIE	31,048	31,022	**57,314**	**41,266**	**−28%**
huff_enc	41	**741***	**445**	**41**	**−91%**
lift	254	43	**1,518**	**57**	**−96%**
matrix1	7	10	**51**	**11**	**−78%**
test3	8,414	8,885	**14,786**	**10,037**	**−32%**

Dom0 Stress Study — Pinning Paradox

Under Dom0 stress, pinning systematically reduces mean & p99 but can worsen WCET:

Benchmark	Δ Mean	Δ p99	Δ Max (WCET)
debie	**−26.5%**	**−35.8%**	**−45.8%**
huff_enc	**−34.1%**	**−31.4%**	**+142.6%**
lift	**−46.3%**	**−37.0%**	**+326.5%**
matrix1	**−30.7%**	**−26.6%**	**−72.1%**
test3	**−15.4%**	**−37.2%**	**−8.2%**

Interpreting the Paradox

- Pinning reduces **variance in the common case** (lower mean/p99 across all 5 benchmarks)
- But if the pinned core is hit by a housekeeping IRQ or Dom0 timer, the interference is **concentrated on one core** instead of distributed
- Result: rare but larger isolated WCET spikes for lift and huff_enc
- Full core isolation (IRQ affinity on all interrupts) would be needed to prevent this

XEN — VIRTUALIZATION TECHNOLOGY COMPARISON

PV & PREEMPT_RT: An Incompatibility, Not a Priority Bug

With the HVM priority-inversion fix confirmed, we extend to guest types with no Device Model at all — PV and PVH

What Happened

- Configuring a PV guest with a PREEMPT_RT kernel reproduced the same failure seen earlier during initial setup — this time, with terminal-routed logs, the cause was identifiable
- Logs revealed a **soft CPU lockup**, pointing directly at the PREEMPT_RT patch itself as the source of the failure
- **Hypothesis:** PREEMPT_RT replaces spinlocks with mutexes at a low level — a change that likely conflicts with how paravirtualization cooperates with the hypervisor
- This also explains why the RT patch fails on **Dom0 itself**: Dom0 is inherently a PV guest

A Second Dead End: PVH Dom0

An attempt to configure Dom0 itself as a PVH guest resulted in **continuous automatic reboots**. With no system logs or graphical output available to diagnose the failure, the root cause could not be identified — this configuration was abandoned.

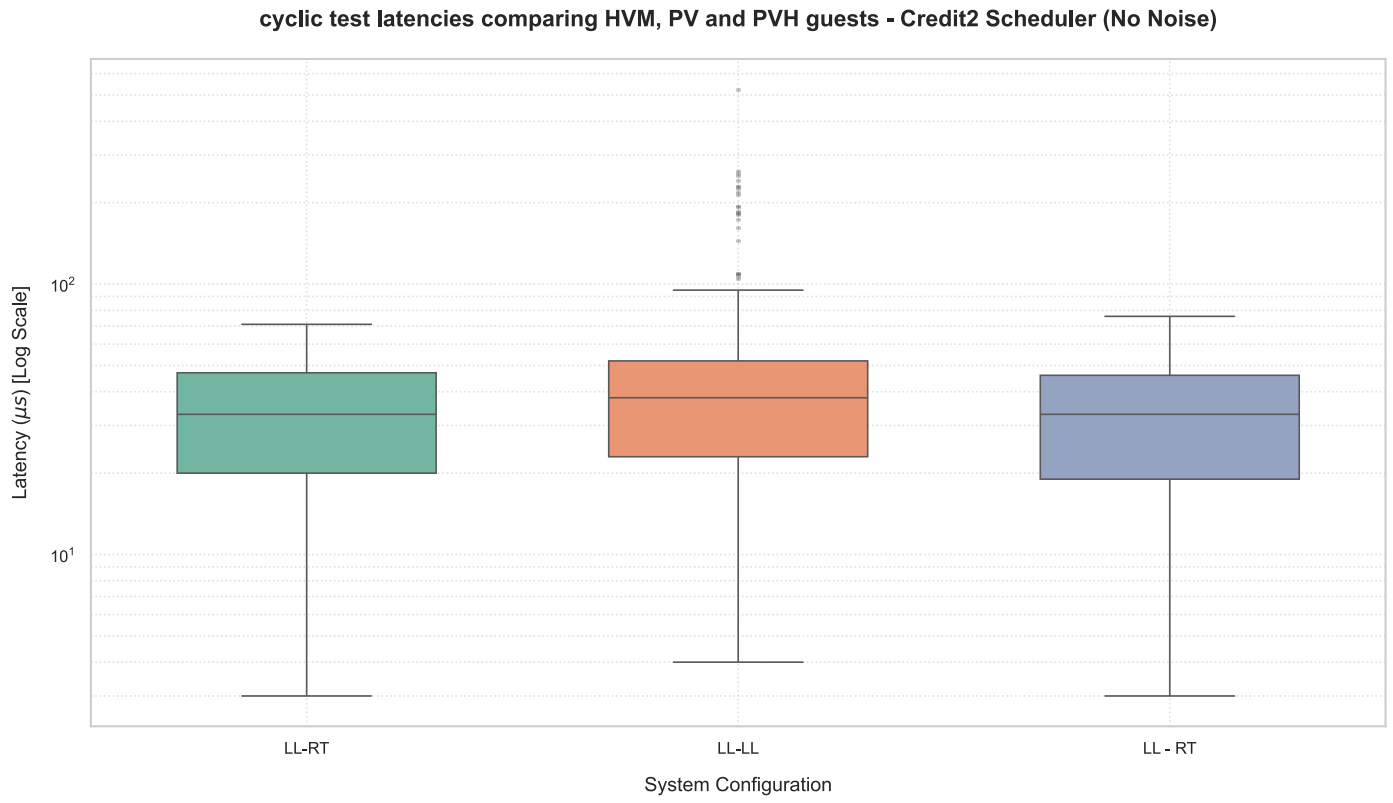
🔧 **Consequence for methodology:** since PREEMPT_RT cannot run on a PV guest, all subsequent PV tests use the **Low-Latency (LL) kernel** on the DomU in place of the RT kernel — PVH remains the only guest type where RT is tested.

❌ PV is therefore structurally excluded from hard real-time use on Xen — not due to a fixable scheduling bug, but a kernel-patch incompatibility at the virtualization layer.

XEN — VIRTUALIZATION TECHNOLOGY COMPARISON

Baseline: LL Dom0, No Stress, No Pinning

PV (LL DomU, forced) vs PVH (RT DomU) — dynamic Credit2 scheduler



Reading the Gap

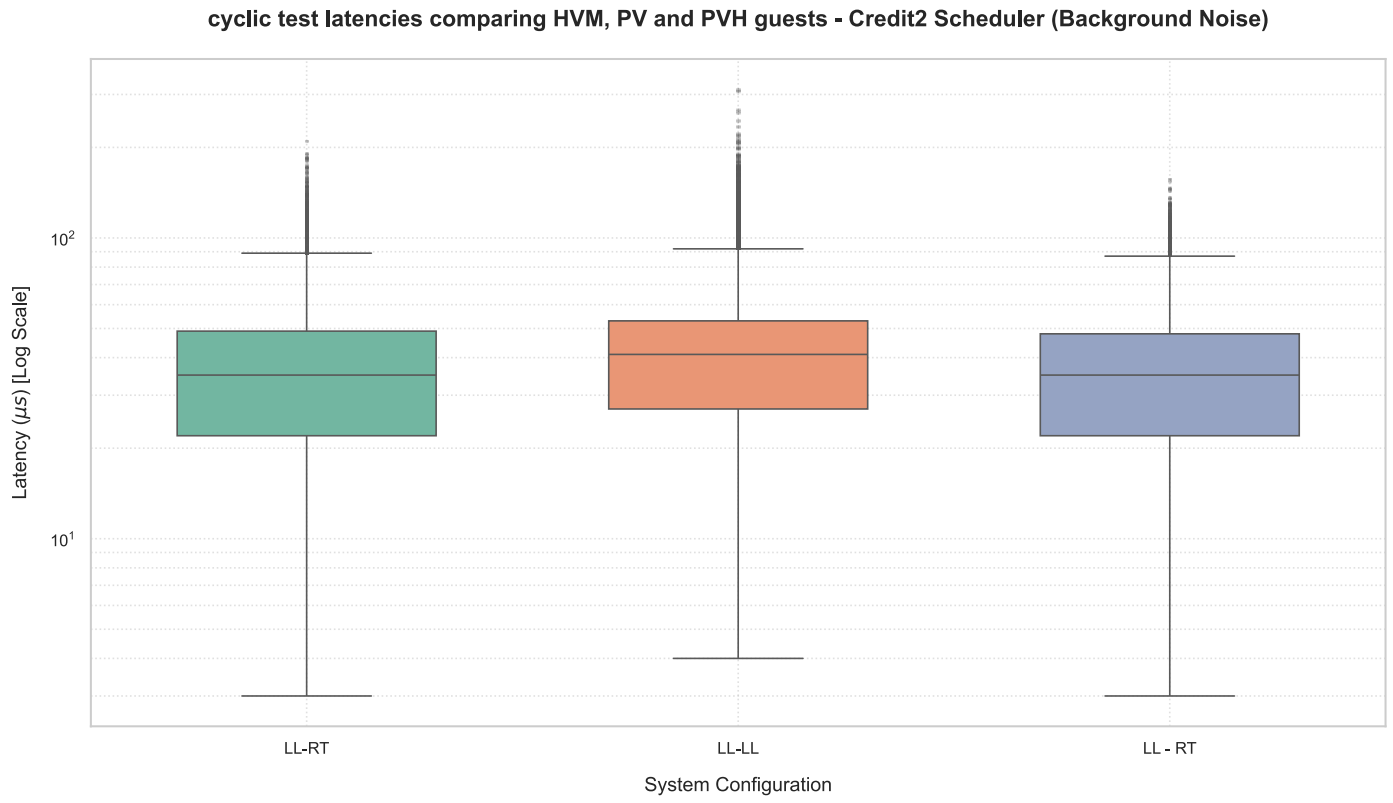
- PV's higher average (37 µs, vs 32 µs for PVH) reflects the baseline overhead of a software-only virtualization path with no hardware-assisted extensions
- The WCET gap is far larger than the average gap — this is a **tail** effect: PV, forced onto the LL kernel, has no RT scheduling guarantees to bound its worst case
- PVH's zero overflows and 76 µs WCET, achieved *without pinning*, show that a genuine RT guest kernel on a lean, DM-free architecture is enough for strong determinism even under dynamic Credit2 scheduling

✅ This baseline gap is the direct, measurable consequence of the PV/PREEMPT_RT incompatibility described on the previous slide — not a scheduling artifact.

XEN — VIRTUALIZATION TECHNOLOGY COMPARISON

PV vs PVH vs HVM Under Dom0 Stress

Credit2 scheduler — background stress workload in Dom0, `cyclictest` probe in PV/PVH/HVM guests



Key Observations

Removing the Device Model isolates PV/PVH behaviour from QEMU-induced preemption — but PV pays for it: **PREEMPT_RT incompatibility** forces a Low Latency kernel, moving the bottleneck from the hypervisor to guest-level scheduling.

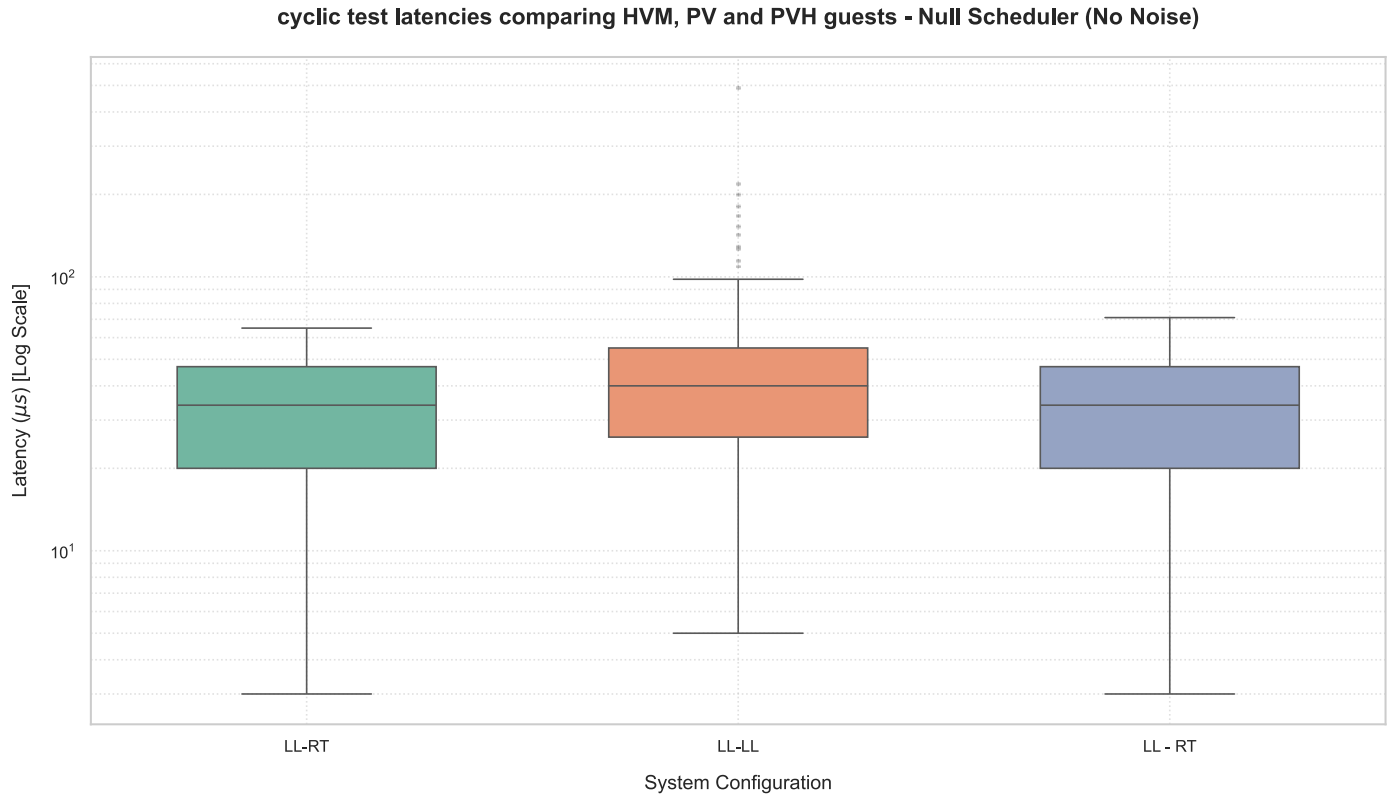
PVH ≈ HVM under contention — both stay within predictable bounds, confirming PVH as a lightweight alternative with full RT resilience.

PV is the worst performer of the three technologies: the lack of RT guest optimizations outweighs any benefit from its lighter virtualization footprint.

XEN — VIRTUALIZATION TECHNOLOGY COMPARISON

PV vs PVH vs HVM Under Static Allocation: Baseline

Strict vCPU-to-pCPU pinning on Dom0 and DomU — emulating an offline NULL scheduler, unstressed conditions



Key Observations

✅ **PVH baseline (71 µs) ≈ HVM (65 µs)** even without stress — static isolation confirms the architectural nuance is minor between the two.

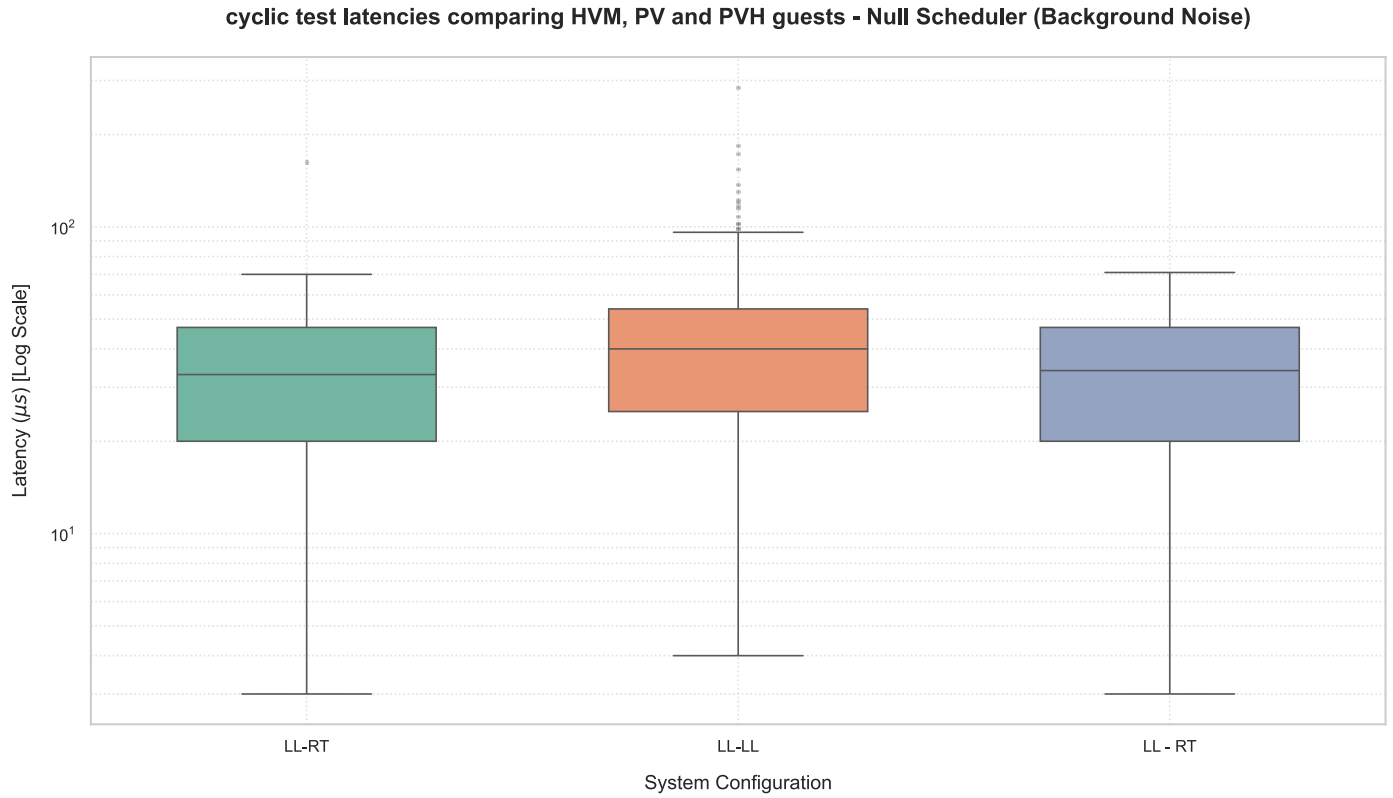
❌ **Pinning alone cannot fix PV:** even under ideal, unstressed hardware isolation, PV's WCET (492 µs) remains ~7× worse than HVM/PVH — the penalty comes from the `PREEMPT_RT`-incompatible Low Latency kernel, not from host contention.

ℹ️ This clean baseline isolates the *inherent* virtualization overhead — any degradation observed once Dom0 stress is introduced can be attributed specifically to contention, not to scheduling variability.

XEN — VIRTUALIZATION TECHNOLOGY COMPARISON

PV vs PVH vs HVM Under Static Allocation + Dom0 Stress

Severe `stressdom0` background workload in the privileged domain, static vCPU pinning retained on both PV and PVH DomUs



Key Observations

⭐ **PVH + Pinning + PREEMPT_RT under stress: 71 µs WCET** — *identical* to its own unstressed baseline. Static isolation successfully shields PVH's critical execution paths, achieving perfect WCET stability regardless of host load.

❌ **PV remains susceptible to host contention** even with static pinning: WCET lowers from 492 µs to a still-measurable interference pattern, confirming that hardware isolation alone cannot compensate for the PREEMPT_RT-incompatible Low Latency kernel.

ℹ️ PVH's isolation from the Device Model plus RT-kernel compatibility lets it fully absorb Dom0 saturation, while PV's guest-level scheduling limitations remain exposed regardless of pinning strategy.

XEN — VIRTUALIZATION TECHNOLOGY COMPARISON

PV vs PVH vs HVM: Summary

PV (Paravirtualization)

- No Device Model — no QEMU in Dom0
- **PREEMPT_RT incompatible** — forced fallback to Low Latency kernel
- Bottleneck shifts from hypervisor to guest-level scheduling
- Lighter footprint cannot compensate for missing RT optimizations

HVM (Full Virtualization)

- QEMU Device Model in Dom0
- PREEMPT_RT compatible
- Historical priority-inversion bug: confirmed FIXED
- Manages latency bounds well under contention

PVH

- No Device Model, no QEMU dependency
- PREEMPT_RT compatible
- Isolated from Device Model interference by design
- Resilience to contention on par with HVM

Baseline WCET Comparison

Configuration	DomU Kernel	Min	Avg ± SD	WCET	Overflows
PV	Low-Latency	4 µs	37 ± 15 µs	525 µs	0
PVH	Real-Time (PREEMPT_RT)	3 µs	32 ± 15 µs	76 µs	0

XEN — VIRTUALIZATION TECHNOLOGY COMPARISON

PV vs PVH vs HVM: Summary

Worst-Case Latencies Under Stress

Config	Stress Average	Stress WCET	Average Δ%	WCET Δ%
PV	40 ± 16 µs	312 µs	+8.16%	-40.57%
PVH	36 ± 15.5 µs	157 µs	+9.82%	+106.58%

Baseline Pinned WCET Comparison

Configuration	DomU Kernel	Min	Avg ± SD	WCET	Overflows
PV	Low-Latency	5 µs	40 ± 15 µs	492 µs	0
PVH	Real-Time (PREEMPT_RT)	3 µs	33 ± 15 µs	71 µs	0

Worst-Case Latencies Under Stress (Pinned)

Config	Stress Average	Stress WCET	Average Δ%	WCET Δ%
PV	39 ± 15 µs	285 µs	-2.13%	-42%
PVH	33 ± 15 µs	71 µs	+0.14%	0%

05

Conclusions & Takeaways

What did we learn, and what does it mean for real-time hypervisor design?

CONCLUSIONS

KVM vs Xen: Head-to-Head Summary

Metric	KVM (RT-RT)	Xen (LL-RT + Pinning)
Baseline WCET	81 µs	65 µs (HVM) / 71 µs (PVH)
Under Stress WCET	~9,800 µs (283 overflows)	163 µs (HVM) / 71 µs (PVH pinned)
Stress + Isolation	~1,935 µs (14 overflows)	71 µs, 0 overflows (PVH)
Scheduling Overhead	~8 µs avg (hypervisor-transparent)	~32–35 µs avg (structural Xen constant)
TACLe (lift, pinned)	70 µs	57 µs
Hard RT Feasibility	Borderline — stress cannot be fully contained	Yes — with LL/RT + vCPU pinning + PVH
Priority Inversion	N/A	Fixed in Xen 4.17.3

⚠️ **KVM advantage:** Much lower average latency overhead (~8 µs vs ~32 µs). Better for soft-RT applications where average matters more than WCET.

✅ **Xen advantage:** Dramatically better WCET containment under stress. PVH + pinning achieves provably stable bounds even with heavy colocated workloads.

CONCLUSIONS

Key Takeaways

① **Determinism is an end-to-end property.** Optimizing only the host OR the guest is insufficient — and can make things worse. The NRT-RT anomaly (6.1 ms WCET!) is a concrete demonstration.

② **vCPU pinning is the dominant factor for Xen.** It reduces overflows from 179 to 48 across the 32-scenario stressor sweep — more impactful than the Dom0 kernel choice. But it is not universally beneficial for WCET under concentrated interference.

③ **The Xen QEMU priority-inversion bug is fixed in Xen 4.17.3.** Elevating QEMU to SCHED_FIFO priority 99 yields no improvement. Modern HVM decouples timer/interrupt delivery from the Device Model process.

④ **PVH + PREEMPT_RT + Pinning is Xen's optimal configuration.** 71 µs WCET even under Dom0 stress — *identical* to its own unstressed baseline. Zero overflows. The PVH architecture removes the Device Model bottleneck while supporting full RT patches.

⑤ **KVM cannot fully contain hardware-interrupt-induced latency spikes** under heavy colocated load — even with CPU isolation (14 overflows, 1.9 ms WCET remain). Xen's Type-1 architecture provides fundamentally better temporal isolation for hard-RT workloads.

CONCLUSIONS

Open Questions & Future Work

Memory Bandwidth Partitioning

Intel RDT / Memguard could further constrain cache-induced latency spikes (cache stressor was most dangerous). Partially explored but needs deeper evaluation.

ARM / RISC-V Platforms

x86 VT-x behaviour differs from ARM interrupt handling. How do these findings translate to embedded platforms (RPi, Jetson)?

Statistical WCET Estimation

5-minute runs may miss rare events. Extreme Value Theory (EVT) / pWCET estimation would strengthen the analysis with probabilistic bounds.

Unikraft Unikernels

Unikraft unikernels as alternative guests — minimal scheduler, no kernel bloat. Promising for extremely lightweight real-time guests but out of scope for final analysis.

Acknowledged Limitations

- Single physical machine — hardware-specific results (AMD CPU, MSI board)
- Single run per configuration — no statistical repetition for TACLe and stressor studies
- NULL scheduler unavailable in Xen 4.17.3 — vCPU pinning used as approximation
- PV analysis limited by PREEMPT_RT incompatibility — LL kernel used as fallback
- Unikraft comparison not completed for final submission

💡 Good research raises more questions than it answers. These limitations are opportunities for future investigation — not fundamental flaws in the methodology.

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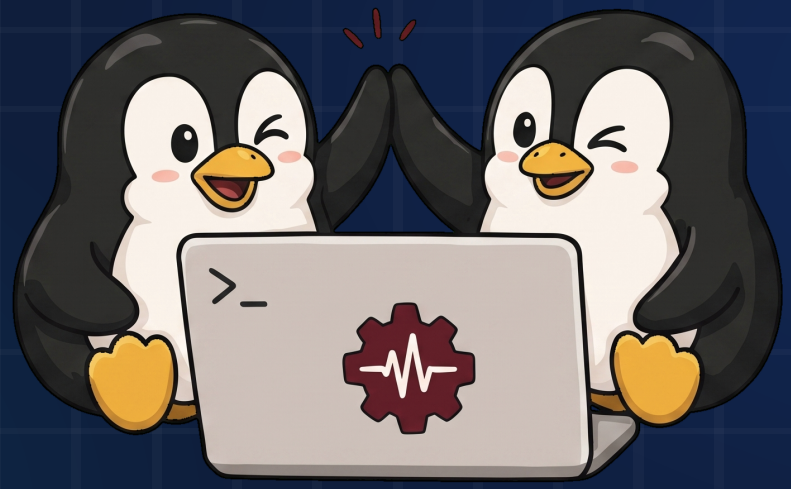
THANK YOU FOR YOUR ATTENTION. QUESTIONS?



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REPOSITORY

github.com/Rita5351/Xen-vs-KVM